

PROJECTION-TRANSPORT STEERING: WHAT THE DECISION BUYS IN CONDITIONAL ACTI- VATION STEERING

Anonymous authors

Paper under double-blind review

ABSTRACT

Conditional activation steering edits a language model’s residual stream only on inputs that a decision flags, and it is usually shown to beat global steering at removing a behavior while sparing the rest. We introduce Projection-Transport Steering (PTS), which separates a conditional edit into a decision and an action, characterizes the minimal action local to a behavioral subspace as an optimal-transport map, and accounts exactly for each part: selectivity equals the decision’s error rates weighted by the action’s conversion rates on the inputs it flags. The account yields a reference every conditional method should beat, detecting the behavior and overriding its readout, and states when steering can beat it. On answer-level overconfidence, in pre-registered comparisons on two model families and two benchmarks, PTS is the most selective method in one setting and level with or below the best in the other two. The account explains each outcome: the decision determines what a gated edit can reach, and detect-and-override with the same decision matches or exceeds the best gated edit in all three.

1 INTRODUCTION

Language models carry many behaviors along directions of their activations, and activation steering changes a behavior by moving the activation along such a direction (Turner et al., 2023; Panickssery et al., 2024; Zou et al., 2023; Li et al., 2023; Arditi et al., 2024). The edit costs one vector operation per token and is applied to every input, so a dose that removes a behavior also changes the inputs that never showed it. Conditional methods place a decision in front of the edit: CAST (Lee et al., 2024) steers only when the prompt resembles a condition vector, and PCHI (Li et al., 2026) rescales attention heads only when a probe flags a wrong but confident answer. Both are evaluated against global steering, and both improve on it.

Such comparisons do not say what the improvement comes from. A conditional edit makes two choices, which inputs to edit and how to edit them, and a better result can come from either. We study overconfidence in multiple-choice reasoning, where the answers to change are the ones the model gets wrong while stating that it is right, and the answers to protect are the ones it gets right and says so. The additive edit that removes every overconfident-wrong answer of `gemma-2-2b-it` also removes every confident-right one (Sec. 5.2), so the problem calls for a decision, and the question is how much the decision, as opposed to the edit, is doing.

We introduce Projection-Transport Steering (PTS), which treats the two choices separately. The action is the smallest edit that moves a projection of the activation to a target law and leaves its orthogonal complement unchanged; it is an optimal-transport map, and shifts, ablation and clamps are its instances. The decision is a likelihood-ratio test on a statistic of the input. Our analysis rests on an identity: the selectivity of any conditional edit equals the decision’s true-positive rate times the rate at which the action converts the flagged target answers, minus its false-positive rate times the rate at which it converts the flagged protected ones. The identity makes the contribution of each choice measurable, and it supplies the reference a conditional edit must beat: flag the behavior with the same decision and override its readout, which converts every flagged answer and edits nothing.

Our main contributions are:

method	transfer function $S(p)$	decision (reads)	cost / token
ActAdd, CAA (Turner et al., 2023; Panickssery et al., 2024)	$p + \alpha$	—	$\Theta(d)$
ITI (Li et al., 2023)	$p + \alpha$ on probe-selected heads	—	$\Theta(d)$
directional ablation (Arditi et al., 2024)	0	—	$\Theta(d)$
SAE clamping (Templeton et al., 2024)	c	—	$\Theta(d d_{\text{SAE}})$
MiMiC (Singh et al., 2024)	full-space affine map	—	$\Theta(d^2)$
Linear-AcT (Rodriguez et al., 2025a)	per-neuron affine map	—	$\Theta(d)$
CAST (Lee et al., 2024)	$p + \alpha$	cosine to a condition vector (prompt)	$\Theta(d)$
PCHI (Li et al., 2026)	learned head rescaling	probe (confidence template)	$\Theta(d)$
PTS (ours)	transport map to a target law	likelihood-ratio probe (prompt, prefix or trace)	$\Theta(d)$

Table 1: **Steering operators as a transfer function plus a decision.** Each operator applies a transfer function S to a projection p of the residual stream; the conditional ones add a decision. For $k = 1$, Theorem 1 identifies the cost-minimal S for a target law, and Proposition 1 gives the selectivity of any row as a function of its decision and its action.

- A selectivity identity for conditional edits, the condition under which steering is more selective than detect-and-override, a corrected Neyman–Pearson characterization of the optimal decision, and the characterization of projection-local actions as transport maps (Sec. 3).
- A pre-registered comparison in which every method is tuned under one rule and evaluated once on unseen questions, for two model families and a second benchmark. PTS is the most selective method on Gemma-MMLU, significantly so against seven baselines, level with the best methods on Qwen-MMLU, and below the best two, not significantly, on Gemma-ARC (Sec. 5.2).
- Evidence that the decision determines the outcome. Across five decisions, the override reference is ordered by the decision’s held-out AUROC, and gated edits reach it only when the action converts nearly every flagged answer; on Qwen the override is more than twice as selective as any gated edit (Sec. 5.3).
- Evidence about where the edit acts: a shift acts mainly in the turn where the model states its confidence and ablation mainly in the reasoning trace (Sec. 5.4); the decision can be taken from the prompt, so the operator runs in a single pass on Gemma (Sec. 5.5).

2 RELATED WORK

Global steering. ActAdd (Turner et al., 2023) and CAA (Panickssery et al., 2024) add a contrastive direction to the residual stream, ITI (Li et al., 2023) shifts probe-selected attention heads, RepE (Zou et al., 2023) reads and writes representation directions, directional ablation (Arditi et al., 2024) projects a direction out, and SAE clamping (Templeton et al., 2024) pins a feature. The same edit is applied to every input, and AxBench (Wu et al., 2025) and studies of steering reliability (Tan et al., 2024; Braun et al., 2025) report that such edits are brittle across inputs. Table 1 places these operators in one family of transfer functions.

Distribution matching. MiMiC (Singh et al., 2024) derives the Gaussian optimal-transport affine map between two activation distributions in the full space, Linear-AcT (Rodriguez et al., 2025a) transports each neuron with a univariate map and a strength parameter, and LinEAS (Rodriguez et al., 2025b) learns affine maps end to end. COAST (Nguyen et al., 2026) characterises the covariance-weighted projection that minimises collateral damage for a point target, and A-LQR (Skifstad et al., 2026) treats steering as linear-quadratic control. These results concern the action. Theorem 1 is the projection-local version of the same transport argument, and we do not claim it as the source of the empirical gains; what is new is its pairing with an exact account of what a decision adds.

Conditional steering. CAST (Lee et al., 2024) applies a behavior vector only when the prompt’s hidden state is similar to a condition vector, and is the baseline closest to our design; we reimplement its prompt-level condition and give it the same labelled data as our detector. Dynamic and semantics-adaptive methods scale the edit per token (Ferrando et al., 2026; Wang et al., 2025; Scalena et al., 2024), and GAPS (Haider et al., 2026) gates individual dimensions. Closest in aim is PCHI (Li et al., 2026), which also separates detection from intervention for verbalised overconfidence: a linear probe on the confidence template conditions a learned rescaling of attention heads, reaching large wrong-confident corrections at small damage on mathematics questions. PCHI learns the action for a fixed readout position; we keep the action fixed and ask what the decision contributes, when it can be taken, and how much it can buy. The accounting identity and ceiling of Sec. 3 apply to PCHI as well. Probes read after a few generated tokens predict whether steering will succeed (Fan et al., 2026), detectors of a finished behavior forecast it worse from a prefix (Kortukov et al., 2026),

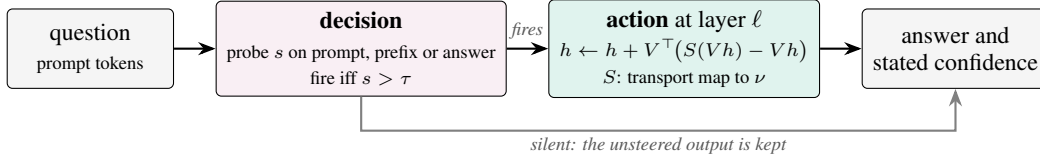


Figure 1: **PTS separates when to edit from how to edit.** A probe reads the input and decides whether to act; the action moves the projection Vh of the residual stream to a target law and leaves the orthogonal complement unchanged. The selectivity of the combination is the decision’s error rates weighted by the action’s conversion rates of the two states (Prop. 1).

and reading a direction off already-edited activations degrades it (Luo et al., 2026); we measure the prefix cost directly and place the reader above the actor. Our detect-and-override reference is selective prediction with a correctness probe: that a model’s activations predict whether its answer is right is well established (Kadavath et al., 2022; Azaria & Mitchell, 2023), and the identity makes it the baseline a conditional edit of a confidence readout has to beat.

3 PROJECTION-TRANSPORT STEERING

A conditional edit makes two choices: how to change an activation, and on which inputs to change it. We call the first the *action* and the second the *decision*. This section fixes the action by a displacement criterion (Sec. 3.1), writes selectivity as a function of the two choices (Sec. 3.2), and describes how the decision is taken in a single pass (Sec. 3.3). Proofs are in Appendix C.

Setup. Let $h \in \mathbb{R}^d$ be the residual-stream activation at a fixed layer and $V \in \mathbb{R}^{k \times d}$ a matrix with orthonormal rows that spans a behavioral subspace. We write $q = Vh$ for the projection, μ for the law of h under unsteered generation, μ_V for the law of q , and ν for a target law on $\mathbb{R}^k$. In our experiments $k = 1$ and $V = \hat{v}^\top$ is the confidence direction, the difference of mean activations between confident and nonconfident answers on an extraction split (Sec. 4).

3.1 THE ACTION

Large edits are what damage generation (Appendix H), so we ask for the smallest edit that moves the projection to a given target law and leaves the rest of the activation alone.

Definition 1 (projection-local steering). *An edit $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is local to V if $(I - V^\top V)T(h) = (I - V^\top V)h$, and it reaches ν if Vh' is distributed as ν when $h \sim \mu$ and $h' = T(h)$. Its cost is $C(T) = \mathbb{E}_\mu \|T(h) - h\|^2$.*

Theorem 1 (local steering is transport). *Let μ_V be absolutely continuous with finite second moment, and let ν have finite second moment. Among local edits that reach ν , the cost is minimised by*

$$T(h) = h + V^\top (S(Vh) - Vh), \quad (1)$$

where S is the optimal-transport (Brenier) map from μ_V to ν ; the minimum cost is $W_2^2(\mu_V, \nu)$, and every minimiser agrees with Eq. (1) on the projection.

Corollary 1 (what a transfer function can and cannot do). *(a) The shift $S(p) = p + \alpha$ is the optimal map iff ν is a translate of μ_V . (b) If two groups of inputs have the same law of p and the event that an input changes state depends on h only through p , every transfer function changes both groups at the same rate.*

In words, once the subspace and the target are fixed, the only freedom left is the one-dimensional map S , and the cheapest S is the monotone rearrangement of quantiles. Table 1 lists the operators this covers. Directional ablation is the map onto δ_0 , a clamp is the map onto a truncated law, and the additive shift $S(p) = p + \alpha$ is the map onto a shifted copy of μ_V . A shift is optimal exactly when the target is a pure translation (Cor. 1); it is the member we end up selecting, and the theorem says it spends no displacement off the confidence coordinate.

3.2 THE DECISION

A decision reads a statistic s of the input and runs the action on the event $\Gamma = \{s > \tau\}$; elsewhere the output is exactly the unsteered one. We measure the result with selectivity, $\text{Sel} = P(\text{leave OCW} \mid$

OCW) $- P(\text{leave } CR \mid CR)$, the share of overconfident-wrong answers (OCW) the edit removes minus the share of confident-right answers (CR) it destroys. Write TPR and FPR for the rates at which the decision fires on OCW and CR, and ρ_O, ρ_C for the rates at which the action changes the state of a flagged OCW and a flagged CR answer.

Proposition 1 (selectivity accounting). *For any decision and action that leave unflagged outputs unchanged,*

$$\text{Sel} = \text{TPR} \cdot \rho_O - \text{FPR} \cdot \rho_C, \quad (2)$$

where ρ_O and ρ_C are conversion rates among flagged answers.

Eq. (2) is an identity, and it separates the two choices. The decision sets (TPR, FPR) and, through which answers it flags, the conversion rates the action achieves on them. Two consequences follow. First, if the action converts flagged and unflagged answers of the same state at the same rate, ρ_O and ρ_C can be read off one ungated run and the equation predicts the selectivity of every decision placed in front of that action. Second, the identity contains its own reference point. A decision that simply overrides the reported confidence of the answers it flags converts every one of them, $\rho_O = \rho_C = 1$, and reaches $\text{TPR} - \text{FPR}$. A gated edit is more selective than this *detect-and-override* reference exactly when

$$\text{TPR}(1 - \rho_O) < \text{FPR}(1 - \rho_C), \quad (3)$$

that is, when the action spares the confident-right answers the decision flags by mistake more often than it fails on the overconfident-wrong answers it flags correctly. The decision term is bounded by how well s separates the two states.

Theorem 2 (optimal decision and its ceiling). *Suppose that within each state the action’s conversion probability does not depend on s . (a) For any objective increasing in TPR and decreasing in FPR, including Eq. (2), thresholding the likelihood ratio of s between OCW and CR is optimal at every operating point. Without the assumption, the optimal decision thresholds the conversion-weighted ratio $p_O(s)r_O(s)/(p_C(s)r_C(s))$, where p is the class-conditional density of s and $r(s)$ the conversion probability given s . (b) If the concave ROC of s has area AUC, then $\max_{\tau}(\text{TPR} - \text{FPR}) \leq 2 \text{AUC} - 1$, so the detect-and-override reference is bounded by $2 \text{AUC} - 1$, and so is any gated edit with $\rho_O = \rho_C$.*

We therefore implement the decision as a logistic-regression probe, an estimate of the log-likelihood ratio, fitted on answers labelled by the evaluation readout, and we judge it by its AUROC on questions it was not fitted to. The same bound explains why the decision should not read the coordinate the action edits. On held-out questions the confidence projection separates OCW from CR at AUROC 0.64 for Gemma and 0.53 for Qwen, so a threshold on it buys little, and an edit that depends only on that coordinate changes both states at rates that differ only through the rest of the activation (Cor. 1).

3.3 DECIDING IN ONE PASS

A decision that reads the finished answer requires generating every flagged question twice. The decision can instead read the prompt, or the mean activation over the first t generated tokens, with a probe fitted on exactly that statistic. Theorem 2 applies to any such statistic, so the cost of deciding early is the AUROC lost relative to the finished trace. The reader sits at or above the layer the action writes, the action starts at the decision token, and positions written before it keep their cached keys. The operator then costs two d -dimensional dot products per token.

4 EXPERIMENTAL SETUP

Models and task. The main model is gemma-2-2b-it (Gemma Team, 2024); the confirmatory comparison is repeated on Qwen2.5-7B-Instruct, a second model family, with its own directions, detector and tuning. Both reason inside `<think>...</think>` and answer `\boxed{letter}` on MMLU (Hendrycks et al., 2021); decoding is greedy. The action is applied at layer 14 of 26 (Gemma) and 14 of 28 (Qwen). A replication on ARC-Challenge (Clark et al., 2018) reuses the Gemma directions, refits every decision on ARC training questions, and follows the same protocol with its own tuning and confirmatory splits (Appendix A).

Splits. All splits are disjoint sets of MMLU test questions (ids in the released code): an *extraction* split of 400 questions for the directions and transport targets, a *detector* pool of 4,000 for the detector, a *tuning* split of 1,200 on which every method’s configuration is chosen, and a *confirmatory* split of 3,000 used once, under a protocol frozen before its steered rollouts were generated (Appendix A).

What is measured. The primary readout (M5) scores the answer the model actually generates. Correctness is the `\boxed{ }` letter against the gold answer; confidence is $P(\text{YES})$ when the model is then asked, in the same conversation, whether that answer is correct. A steered question is steered for both the reasoning trace and the confidence query, so the answer and its confidence come from the same edited model. Each question is in one of four states: *overconfident-wrong* (OCW, wrong with $P(\text{YES}) \geq 0.5$), *nonconfident-wrong*, *nonconfident-right*, and *confident-right* (CR). Two secondary readouts, the letter-logit confidence of the same answer and a per-option verification readout, are reported in Appendix G; the prompts are in Appendix L.

Metrics. Pairing each question with itself before and after the edit, removal is $P(\text{not OCW after} \mid \text{OCW before})$ over the n_{OCW} questions that started in OCW, retention is $P(\text{CR after} \mid \text{CR before})$ over the n_{CR} that started in CR, and selectivity is $\text{Sel} = \text{removal} - (1 - \text{retention})$: 1 for an edit that removes every OCW and keeps every CR, 0 for one that treats them alike. Δacc is the change in accuracy over all questions, and ECE is the 10-bin expected calibration error of the confidence.

Methods and tuning. We compare prompting (a calibration instruction), additive CAA (Panickssery et al., 2024), directional ablation (Arditi et al., 2024), MiMiC (Singh et al., 2024), Linear-Act (Rodriguez et al., 2025a) and two conditional shifts: CAST with its original prompt condition, a cosine threshold to a condition vector computed from the same labelled pool our detector uses (Lee et al., 2024), and a CAST-style variant conditioned on the finished trace’s projection onto the extraction-split OCW–CR direction. All are reimplemented on the same hooks, records and readouts. The PTS decision is a logistic-regression probe on the mean activation of one layer, fitted on the detector pool, with its layer and regularisation chosen by AUROC on the tuning split; the action is ablation or a shift of the confidence coordinate. Each method searches a grid of 8 to 16 configurations on the tuning split, and one rule selects all of them: the highest selectivity among configurations whose Δacc is at least -0.01 (Appendix E). Post-hoc PTS reads the finished unsteered answer and regenerates the flagged questions; single-pass PTS decides at the prompt, or after $t \in \{16, 32, 64, 128\}$ generated tokens with a probe fitted on that prefix, and acts from then on.

Statistics. The unit of analysis is the question. Intervals are question-level paired bootstrap intervals with 10^4 resamples; comparisons between methods resample the same questions for both. The seven primary comparisons of post-hoc PTS against the baselines are Holm-adjusted. We call an accuracy change non-inferior when the lower end of its 95% interval is above -0.02 , a margin fixed in the protocol.

5 RESULTS

5.1 THE SELECTIVITY IDENTITY PREDICTS GATED EDITS

The identity turns one ungated run into a prediction for every decision. On the Gemma tuning split we measure the conversion rates of six actions once, compose each with two decisions at 20 thresholds, and compare the prediction $\text{TPR } \rho_O - \text{FPR } \rho_C$ with the measured selectivity (Fig. 2a). The two agree with correlation 0.97 over 240 combinations, and with correlation 0.91 over the 240 Qwen combinations, whose selectivities span a narrower range; the prediction is higher by 0.033 on average for Gemma and by 0.012 for Qwen. The bias comes from the independence assumption: the answers a decision flags are converted more often than the average answer of their state, and more so in the protected state. With the flagged conversion rates in place of the ungated ones the identity is exact. Because a post-hoc gated arm is, question by question, either its unsteered or its ungated output, the gated selectivity can be composed from one ungated run; on the confirmatory split the four gated arms land within 0.010 of that composition (Appendix F), which is how we score gated configurations during tuning.

The identity also says what each choice contributes. The shift the selection rule picks for Gemma, $S(p) = p - 0.375\|h\|$, changes the state of 84% of overconfident-wrong and 59% of confident-right answers when applied everywhere. By Prop. 1, a decision recovers 0.59 for every confident-right

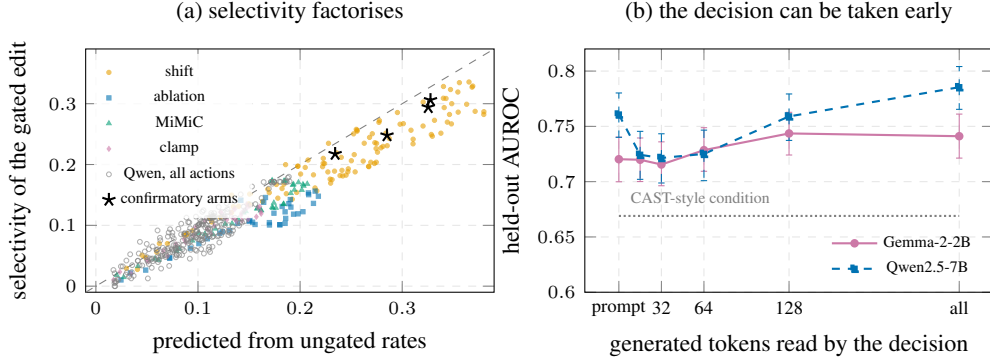


Figure 2: **The decision is the lever, and it can be taken early.** (a) Each point is a decision–action–threshold combination on the tuning split (240 for Gemma in colour, 240 for Qwen in grey). The horizontal axis is $\text{TPR } \rho_O - \text{FPR } \rho_C$ with ungated conversion rates; the vertical axis is the selectivity of the gated edit, composed from the same ungated run. Stars are the four gated arms of the confirmatory split, measured on real gated runs. (b) AUROC between OCW and CR on the 3,000 confirmatory questions of a probe that reads the prompt, the first t generated tokens, or the whole answer; bars are 95% bootstrap intervals, and the dotted line is the trace direction used by the CAST-style condition.

method	Gemma-2-2B, MMLU ($n_{OCW}=963, n_{CR}=1,399$)				Qwen2.5-7B, MMLU ($n_{OCW}=563, n_{CR}=2,007$)				Gemma-2-2B, ARC ($n_{OCW}=228, n_{CR}=780$)			
	Δacc	ret.	Sel $\uparrow$	$\Delta\text{Sel} (p)$	Δacc	ret.	Sel $\uparrow$	$\Delta\text{Sel} (p)$	Δacc	ret.	Sel $\uparrow$	$\Delta\text{Sel} (p)$
prompting	+0.005	0.78	0.200 [0.16, 0.24]	−0.094 (<0.001)	−0.003	0.89	0.212 [0.17, 0.25]	+0.040 (0.313)	+0.003	0.238 [0.17, 0.31]	+0.003 (1.000)	
additive CAA	−0.001	0.66	0.220 [0.18, 0.26]	−0.074 (0.008)	−0.015 [†]	0.92	0.151 [0.12, 0.19]	−0.021 (0.998)	−0.001 [†]	0.294 [0.22, 0.37]	+0.059 (0.356)	
dir. ablation	+0.009	0.72	0.173 [0.13, 0.21]	−0.121 (<0.001)	−0.009 [†]	0.92	0.156 [0.12, 0.19]	−0.016 (0.998)	−0.008 [†]	0.215 [0.14, 0.29]	−0.020 (1.000)	
MiMiC	+0.000	0.80	0.117 [0.08, 0.15]	−0.177 (<0.001)	−0.002	0.92	0.154 [0.12, 0.19]	−0.018 (0.998)	+0.010	0.198 [0.13, 0.26]	−0.037 (1.000)	
Linear-AcT	+0.005	0.92	0.061 [0.04, 0.09]	−0.233 (<0.001)	−0.003	0.94	0.123 [0.09, 0.16]	−0.049 (0.133)	+0.003	0.109 [0.06, 0.16]	−0.126 (0.008)	
CAST (prompt)	−0.002	0.76	0.217 [0.18, 0.26]	−0.077 (0.004)	−0.011 [†]	0.92	0.122 [0.09, 0.16]	−0.050 (0.133)	−0.003	0.109 [0.05, 0.17]	−0.126 (0.003)	
CAST-style (trace)	−0.005	0.70	0.248 [0.21, 0.29]	−0.046 (0.020)	−0.007	0.98	0.040 [0.02, 0.06]	−0.132 (<0.001)	+0.001	0.289 [0.22, 0.36]	+0.054 (0.356)	
PTS, post-hoc	+0.003	0.80	0.294 [0.26, 0.33]	—	−0.002	0.93	0.172 [0.14, 0.21]	—	−0.006 [†]	0.235 [0.17, 0.30]	—	
PTS, prefix	−0.008	0.73	0.294 [0.26, 0.33]	+0.000	−0.008	0.94	0.109 [0.08, 0.14]	−0.063	−0.008 [†]	0.276 [0.21, 0.35]	+0.041	
PTS, prompt	−0.012 [†]	0.76	0.306 [0.27, 0.35]	+0.012	−0.034 [†]	0.89	0.120 [0.08, 0.16]	−0.052	−0.027 [†]	0.182 [0.11, 0.25]	−0.053	

Table 2: **PTS is the most selective method on Gemma-MMLU, level with the best methods on Qwen-MMLU, and below the two best on Gemma-ARC, at unchanged accuracy.** Confirmatory splits of 3,000 MMLU questions per model and the 1,165 ARC-Challenge test questions, each method at the configuration it won on its tuning split under one rule. Retention is over the questions that started confident-right. Sel intervals are question-level bootstrap intervals (10^4 resamples); ΔSel is paired against PTS post-hoc, with Holm-adjusted p over the seven baselines. [†]: the 95% interval of Δacc reaches below -0.02 . Rows below the line are secondary.

answer it leaves alone and forfeits 0.84 for every overconfident-wrong answer it misses, so the value of detection grows with how indiscriminate the action is. The probe reads the unsteered answer at held-out AUROC 0.74, against 0.67 for the direction that conditions the CAST-style baseline (Fig. 2b); at its selected threshold it fires on 54% of overconfident-wrong and 22% of confident-right answers.

5.2 PRE-REGISTERED COMPARISON ON TWO MODEL FAMILIES

A global shift trades one state for the other. On the Gemma tuning split, the shift $p - 0.75\|h\|$ removes every overconfident-wrong answer and every confident-right one, at a cost of 15.3 accuracy points; smaller doses keep more of both (Appendix E). The tuned dose, $p - 0.25\|h\|$, reaches selectivity 0.197 there.

PTS is the most selective method on Gemma. Table 2 reports the frozen configurations on the 3,000 confirmatory questions. PTS reaches selectivity 0.294 [0.256, 0.334], removing 49% of overconfident-wrong answers while keeping 80% of confident-right ones. The strongest baseline, the CAST-style shift conditioned on the trace direction, reaches 0.248, and the paired difference to PTS is 0.046 [0.008, 0.085]; the other six baselines trail by 0.074 to 0.233, and all seven differences remain significant after Holm adjustment. Accuracy changes by +0.003 [−0.009, 0.015], inside the two-point margin, as it does for every baseline. Against the same shift applied to every question, PTS keeps 0.80 instead of 0.42 of the confident-right answers at a similar selectivity (0.294 against 0.256).

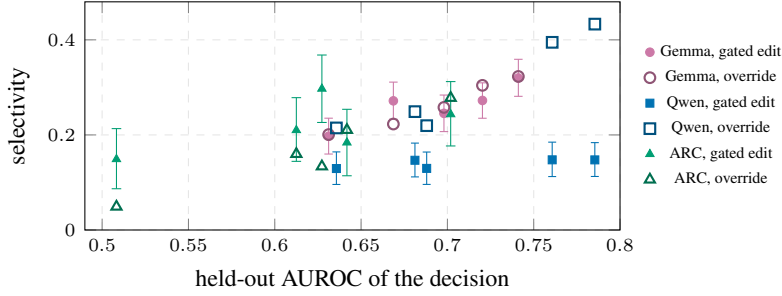


Figure 3: **The override reference is ordered by the held-out quality of the decision, and gated edits reach it only when the action converts what the decision flags.** Each marker pair is one of five decisions (probes on the answer or the prompt, the CAST prompt condition, and a difference-of-means direction on the answer fitted on 400 or 4,400 labelled answers), paired with the action and threshold it selects on the tuning split. Selectivity is measured on the confirmatory split with 95% bootstrap intervals; filled markers steer, hollow markers override the reported confidence of the flagged answers and edit nothing.

On Qwen the identity predicts where the gate helps. On Qwen the tuning split selects ablation as the action, and ablation applied everywhere changes 24% of overconfident-wrong but only 8% of confident-right answers. By Prop. 1, a decision can recover at most that 8% on each answer it leaves alone, so there is little for it to add: the probe raises selectivity from 0.156 for ungated ablation to 0.172 [0.136, 0.210]. PTS is level with prompting (0.212, paired difference -0.040 [$-0.083, 0.004$]), additive steering, ablation and MiMiC, and above Linear-Act and both CAST variants; after Holm adjustment only the difference to the CAST-style variant is significant. Its accuracy change, -0.002 [$-0.014, 0.009$], is inside the margin; the intervals of the tuned additive dose (-0.015 [$-0.027, -0.003$]) and of ablation reach below it.

On ARC the action alone is already selective. The ARC replication selects a milder shift, $p - 0.25\|h\|$, which applied everywhere changes 51% of overconfident-wrong and 22% of confident-right answers. A decision then forfeits more on the target answers it misses than it recovers on the protected answers it spares. PTS reaches 0.235 [0.168, 0.303], below the ungated dose (0.294) and the CAST-style shift (0.289) by differences that are not significant after Holm adjustment, and significantly above Linear-Act and the CAST prompt condition, whose decision is at chance on ARC (AUROC 0.51).

5.3 THE DECISION DECIDES

The override reference is ordered by the decision, and gated edits follow it when the action converts what the decision flags. We add two decisions to the three of the protocol, a difference-of-means direction on the answer fitted on the 4,400 labelled answers of the detector pool, and a detect-and-override reference for each decision that lowers the reported confidence of the flagged answers below 0.5 and edits nothing. Each is selected on the tuning split and evaluated on the confirmatory split by composition, which Appendix F shows to match real gated runs. Fig. 3 shows the result. The override reference is ordered by the held-out AUROC of its decision, as Theorem 2 requires: on Gemma it runs from 0.201 for the CAST prompt condition (AUROC 0.63) to 0.323 [0.284, 0.362] for the probe on the answer (AUROC 0.74). On Gemma the frozen shift converts 91% of flagged overconfident-wrong and 89% of flagged confident-right answers, so by Eq. (3) a gated edit can at best match the override, and it does: the best gated edit reaches 0.320, and the frozen PTS arm 0.294. On Qwen the selected action is ablation, which converts 28% of flagged overconfident-wrong and 15% of flagged confident-right answers, and every gated edit stays between 0.13 and 0.17 while the override with the same probe reaches 0.433 [0.393, 0.472]. On ARC, where the probe reaches AUROC 0.70, the override reaches 0.278 [0.207, 0.348] and the best gated edit 0.297. The override also calibrates better on Qwen: it lowers ECE from 0.216 to 0.139 and Brier score from 0.230 to 0.195, where PTS leaves both unchanged; on Gemma PTS lowers ECE from 0.354 to 0.255 and the override to 0.234. The override analysis was added after the protocol and is not pre-registered.

5.4 WHERE THE EDIT ACTS

Which part of generation the edit changes depends on the action. We reran the frozen PTS decision with the action applied only during the reasoning trace, or only during the turn in which the model states its confidence. On Gemma, where the action is a shift, steering only the confidence turn reaches selectivity 0.274 [0.237, 0.312], close to the 0.294 of the full edit, and steering only the trace reaches 0.159. On Qwen, where the action is ablation, the pattern reverses: the trace-only edit reaches 0.172, equal to the full edit, and the confidence-turn edit 0.006. A shift of the confidence coordinate therefore acts mainly by lowering the stated confidence directly, while ablation acts through the reasoning. The steered traces also change answers: on Gemma PTS turns 124 of the 963 overconfident-wrong answers into correct ones and 106 of the 1,399 confident-right answers into wrong ones (Appendix G).

5.5 DECIDING IN ONE PASS

Most of the detection is available before the answer is written. Fig. 2b plots the held-out AUROC of a probe fitted on each statistic. For Gemma the prompt alone reaches 0.720 and the first 16 tokens 0.720, against 0.741 for the finished answer; for Qwen the prompt reaches 0.761 against 0.785. On Gemma the single-pass operators in Table 2 match the post-hoc one: deciding at token 32 gives selectivity 0.294 and deciding at the prompt 0.306, with paired differences to the post-hoc decision of 0.000 and +0.012. On Qwen the gap in AUROC is larger, and the selected single-pass variants reach 0.109 (token 16) and 0.120 (prompt), below the post-hoc 0.172. A decision at the prompt costs one dot product on an activation the forward pass already computes, so the operator adds no generation; in an end-to-end measurement of the earlier prefix gate the single-pass operator ran at $1.05\times$ the unsteered wall clock, against $1.97\times$ for regenerating flagged answers (Appendix K).

6 CONCLUSION

We have presented Projection-Transport Steering, which splits a conditional activation edit into a decision and an action and accounts exactly for each. The accounting has a practical consequence for how conditional steering is evaluated. A gated edit should be compared not only with global steering but with detect-and-override under the same decision, and the identity says in advance when it can win: only if the action spares the protected answers the decision flags by mistake more often than it misses the target answers it flags correctly. In our experiments the decision determined the outcome, the best decision was a probe trained on held-out labels, and overriding the readout with it was as selective as steering on one model and far more selective on the other.

Our study has limitations. The evidence covers one behavior, measured with one confidence readout, on two benchmarks and two instruction-tuned models; an earlier form of the operator passed on one of three models on TruthfulQA (Appendix B). The decision is a probe trained on labelled answers, and its threshold is tied to the distribution it was fitted on (Appendices H and J). The override reference applies only where the behavior has a readout that can be overwritten; for behaviors expressed in free text, the conversion rates of the action become the whole question.

AI USE STATEMENT

We used generative AI tools to polish the manuscript and to retrieve related work. The authors verified every resulting claim and reference and take responsibility for the final content.

REFERENCES

- Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda, *Refusal in language models is mediated by a single direction*, Advances in Neural Information Processing Systems 37, 2024.
- Amos Azaria and Tom Mitchell, *The internal state of an LLM knows when it’s lying*, Findings of the Association for Computational Linguistics: EMNLP, 2023.

- J. Braun, C. Eickhoff, D. Krueger, S. A. Bahrainian, D. Krashennnikov. *Understanding (un)reliability of steering vectors in language models*. ICLR Workshop on Foundation Models in the Wild, 2025.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord, *Think you have solved question answering? Try ARC, the AI2 reasoning challenge*, arXiv:1803.05457, 2018.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, et al., *Training verifiers to solve math word problems*, arXiv:2110.14168, 2021.
- Chenrui Fan, Yize Cheng, Ming Li, Soheil Feizi, and Tianyi Zhou, *When is your LLM steerable? Predicting steering success from internal states*, arXiv:2606.11599, 2026.
- Javier Ferrando, Xavier Suau, Pau Rodriguez, and Adrià Gonzalez, *Dynamically scaled activation steering*, arXiv:2512.03661, 2026.
- Gemma Team, *Gemma 2: Improving open language models at a practical size*, arXiv:2408.00118, 2024.
- Umair Haider, Hassan Sajjad, and A. B. Siddique, *GAPS: gated activation steering at dimension granularity*, arXiv:2609.01878, 2026.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt, *Measuring massive multitask language understanding*, International Conference on Learning Representations, 2021.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, et al., *Language models (mostly) know what they know*, arXiv:2207.05221, 2022.
- Evgenii Kortukov, Piotr Komorowski, Florian Klein, Paula Engl, Gabriele Sarti, Seong Joon Oh, Sebastian Lapuschkin, and Wojciech Samek, *Predicting future behaviors in reasoning models enables better steering*, arXiv:2606.11172, 2026.
- Bruce W. Lee, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Erik Miehl, Pierre Dognin, Manish Nagireddy, and Amit Dhurandhar, *Programming refusal with conditional activation steering*, International Conference on Learning Representations, 2025.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg, *Inference-time intervention: Eliciting truthful answers from a language model*, Advances in Neural Information Processing Systems 36, 2023.
- Ke Li, Chongzhe Zhang, Zifan Zeng, Feng Liu, Qunli Zhang, and Zheng Hu, *Calibrating overconfidence without sacrificing confidence: Probe-conditioned head intervention for LLMs*, arXiv:2606.09876, 2026.
- Ziyang Luo, Zhongyao Chu, Xinjie He, Youting Wang, Xukui Qin, Runxiong Wu, and Yan-Syuan Chen, *You only pass once: answering and abstaining together in a single forward pass of a frozen language model*, arXiv:2608.14465, 2026.
- Tam Nguyen, Tu Anh Nguyen, Sina Alemohammad, and Richard G. Baraniuk, *Minimizing collateral damage in activation steering*, arXiv:2605.01167, 2026.
- Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner, *Steering Llama 2 via contrastive activation addition*, Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, 2024.
- Pau Rodriguez, Arno Blaas, Michal Klein, Luca Zappella, Nicholas Apostoloff, Marco Cuturi, and Xavier Suau, *Controlling language and diffusion models by transporting activations*, International Conference on Learning Representations, 2025.
- Pau Rodriguez, Michal Klein, Eleonora Gualdoni, Arno Blaas, Luca Zappella, Marco Cuturi, and Xavier Suau, *LinEAS: End-to-end learning of activation steering with a distributional loss*, arXiv:2503.10679, 2025.

- Daniel Scalena, Gabriele Sarti, and Malvina Nissim, *Multi-property steering of language models with dynamic activation composition*, Proceedings of BlackboxNLP, 2024.
- Shashwat Singh, Shauli Ravfogel, Jonathan Herzig, Roei Aharoni, Ryan Cotterell, and Ponnurangam Kumaraguru, *Representation surgery: Theory and practice of affine steering*, Proceedings of the 41st International Conference on Machine Learning, 2024.
- Julian Skifstad, Xinyue Annie Yang, and Glen Chou, *Local Linearity of LLMs Enables Activation Steering via Model-Based Linear Optimal Control*, arXiv:2604.19018, 2026.
- D. Tan, D. Chanin, A. Lynch, B. Paige, D. Kanoulas, A. Garriga-Alonso, R. Kirk. *Analysing the generalisation and reliability of steering vectors*. NeurIPS, 2024.
- Adly Templeton, Tom Conerly, Jonathan Marcus, et al., *Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet*, Transformer Circuits Thread, 2024.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid, *Steering language models with activation engineering*, arXiv:2308.10248, 2023.
- Weixuan Wang, Jingyuan Yang, and Wei Peng, *Semantics-adaptive activation intervention for LLMs via dynamic steering vectors*, International Conference on Learning Representations, 2025.
- Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts, *AxBench: Steering LLMs? Even simple baselines outperform sparse autoencoders*, Proceedings of the 42nd International Conference on Machine Learning, 2025.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, et al., *Representation engineering: A top-down approach to AI transparency*, arXiv:2310.01405, 2023.

A CONFIRMATORY PROTOCOL

The protocol was committed to the repository before any steered rollout on the confirmatory split was generated (rebuttal/protocols/confirmatory-v19.md, with two amendments made after the tuning pass and before launch), together with the SHA-256 hashes of the selected configurations and of every fitted detector (confirmatory-v19-gemma.sha256).

Splits. All splits are subsets of the 14,042-question MMLU test set, identified by index and constructed by `label_pool.split_records`. The extraction split (400) excludes the first evaluation subset of the earlier version; the tuning split (1,200) excludes the extraction split and the five earlier evaluation subsets; the detector pool (4,000) and the confirmatory split (3,000) are consecutive blocks of a seeded permutation of the remaining questions. No question appears in two of the four splits.

Selection rule. For every method the configuration with the highest tuning-split selectivity among those with $\Delta_{\text{acc}} \geq -0.01$ is carried unchanged to the confirmatory split. Gated configurations are scored on the tuning split by combining the ungated rollouts of the same split with the gate’s flags (Prop. 1 makes this exact up to batched-decoding nondeterminism; Appendix F); single-pass configurations are real runs. Every confirmatory number is a real run.

Tests. Primary: PTS post-hoc against each of the seven baselines on pooled M5 selectivity, question-level paired bootstrap with 10^4 resamples, two-sided, Holm-adjusted over the seven comparisons. Accuracy: an arm is non-inferior if the lower end of the 95% interval of its accuracy change is above -0.02 . The single-pass and prompt-decision variants, M4 and M2 selectivities, removal, retention and ECE are secondary and reported without a claim.

Released records. Every unsteered and steered answer of the tuning and confirmatory splits is released with its state, confidence and correctness (`results/release/records.csv.gz`); `paper/recompute_from_records.py` recomputes each selectivity and paired interval in Table 2 from that file alone.

ARC replication. A second protocol (`confirmatory-v20-arc.md`), frozen before any steered ARC rollout, repeats the Gemma comparison on ARC-Challenge: a detector pool of 1,012 and a tuning split of 400 from the ARC training and validation questions, and the 1,165-question test split as the confirmatory split. The directions, moments and action layer are those fitted on MMLU; every decision is refitted on the ARC pool together with the MMLU pool, and every configuration is selected on the ARC tuning split with the same rule and grids.

Analyses added after the protocol. The detect-and-override reference, the difference-of-means decision fitted on 4,400 labels, the decision-by-action matrix (Sec. 5.3), the trace-only and confidence-turn-only edits (Sec. 5.4), calibration and the out-of-domain negatives were run after the confirmatory results were known. They use the same splits and the frozen decisions, and we report them as exploratory.

B AN EARLIER OPERATOR ON FURTHER MODELS

Before the present protocol we froze a simplified operator, an action along a detection direction, after a development analysis on Qwen3, and tested it on held-out TruthfulQA panels of 237 questions (50 in-domain, 187 shifted) for Mistral-7B, OLMo-2-7B and Granite-8B. Each comparison was against a norm-matched spherical action, with 10,000 paired BCa resamples and Bonferroni-adjusted two-sided intervals, and a model passed only if calibrated MC2 and MC1 both improved with intervals excluding zero.

model / panel	comparison	Δ calibrated MC2 [adj. 95% CI]	Δ MC1 [adj. 95% CI]	decision
OLMo-2 / TruthfulQA	matched spherical	+ .137 [+ .105, + .168]	+ .105 [+ .025, + .177]	pass
OLMo-2 / TruthfulQA	no intervention	+ .299 [+ .251, + .345]	+ .287 [+ .177, + .384]	pass
OLMo-2 / TruthfulQA	strongest isotropic	+ .220 [+ .180, + .258]	+ .194 [+ .097, + .287]	pass
OLMo-2 / TruthfulQA	unprompted fit	+ .295 [+ .247, + .343]	+ .287 [+ .177, + .384]	pass

Table 3: Held-out OLMo-2 validation on 237 TruthfulQA questions, 50 in-domain and 187 shifted. Intervals use 10,000 paired BCa resamples and the predeclared Bonferroni family; the table shows the matched action and representative construction controls of the seven-control family, all of which pass.

On OLMo-2 the operator passed against the matched action and against all seven pre-registered construction controls (Table 3). Mistral improved calibrated MC2 by +0.153 [+0.108, +0.197], but its MC1 interval [-0.059, +0.122] included zero, and Granite did not clear the screening threshold. A separate screen of the same three models on BBQ and ETHICS, six model-behavior cells with disjoint fitting, screening and test units, advanced one cell, Mistral on ETHICS, to a 1,000-scenario test. There the operator improved pooled calibrated MC2 by +0.107 [+0.090, +0.124] and lowered MC1 on label-1 scenarios by 0.093 [0.071, 0.124].

C PROOFS AND FORMAL STATEMENTS

Theorem 1. By locality, $T(h) - h \in \text{span}(V^\top)$ μ -a.s., hence $\|T(h) - h\|^2 = \|VT(h) - Vh\|^2$ because the rows of V are orthonormal, and $C(T) = \mathbb{E}\|VT(h) - Vh\|^2$. The joint law γ of $(Vh, VT(h))$ is a coupling of (μ_V, ν) by the goal constraint, so $C(T) = \int \|y - x\|^2 d\gamma \geq W_2^2(\mu_V, \nu)$. Because μ_V is absolutely continuous and both laws have finite second moments, Brenier’s theorem gives a unique optimal coupling, induced by $S = \nabla\varphi$ with φ convex. $T^*(h) = h + V^\top(S(Vh) - Vh)$ is admissible ($VV^\top = I_k$ gives $VT^*(h) = S(Vh)$, and the complement is untouched) and attains the bound. For any minimiser the induced coupling is optimal, hence equal to the unique one, so $VT(h) = S(Vh)$ a.s. $\square$

For $k = 1$ absolute continuity can be weakened to atomlessness of μ_V : the monotone rearrangement $F_V^{-1} \circ F_{\mu_V}$ is then an optimal map for every ν with a finite second moment, including targets with atoms such as δ_0 (ablation) or a law truncated at c (clamp); uniqueness of the map can fail only on a μ_V -null set.

Proposition 2 (metric covariance). *If the cost is $(\Delta h)^\top M(\Delta h)$ with $M \succ 0$, the admissible problem is optimal transport of μ_V to ν in $\mathbb{R}^k$ under the cost $\|\cdot\|_{M_V}^2$, $M_V = VMV^\top \succ 0$, and the optimal operator is $T^*(h) = h + V^\top(S_M(Vh) - Vh)$ with $S_M = M_V^{-1/2} \circ \nabla\varphi \circ M_V^{1/2}$, φ convex.*

Proof. Locality writes $\Delta h = V^\top a$ with $a \in \mathbb{R}^k$, and the goal constraint with $VV^\top = I_k$ fixes $a = S(Vh) - Vh$ for the map S in use. Then $\|\Delta h\|_M^2 = a^\top M_V a$. The cost is a latent-space Mahalanobis cost on $\mathbb{R}^k$; it is not the full-space Mahalanobis distance of Δh unless M commutes with $V^\top V$. In the coordinates $z = M_V^{1/2} q$ it is Euclidean, the change of variables is internal to $\mathbb{R}^k$, and Brenier's theorem in z (the pushforward of an absolutely continuous law under an invertible linear map is absolutely continuous) gives $\nabla \varphi$; mapping back gives S_M . $\square$

Corollary 1. (a) If $\nu = \mu_V * \delta_\alpha$ the map $x \mapsto x + \alpha$ is the gradient of a convex function pushing μ_V to ν , hence the Brenier map. Conversely, if the optimal map is $x \mapsto x + \alpha$ then ν is its pushforward, the shift. For one-dimensional Gaussians the monotone map is $x \mapsto m_t + (\sigma_t/\sigma_s)(x - m_s)$, with slope σ_t/σ_s . (b) Let E be the event that a trace changes state under the edit and suppose $P(E | h) = g(p, S(p))$. For subpopulation A , $P(E | A) = \mathbb{E}[g(p, S(p)) | A] = \int g(p, S(p)) d\mu_p^A$, and the same for B . If $\mu_p^A = \mu_p^B$ the two integrals coincide. $\square$

The premise of (b) is a conditional-independence assumption: given p , the state change must not depend on the complement $h_\perp$ or on the subpopulation. It is the premise, not the conclusion, that the data reject (Sec. 3).

Proposition 1. Let R be the event that an OCW trace leaves OCW and L the event that a CR trace leaves CR. Off Γ the output equals the unsteered output, which is in its original state by definition, so $R \subseteq \Gamma$ and $L \subseteq \Gamma$ up to null sets. Then $P(R | OCW) = P(\Gamma | OCW)P(R | OCW, \Gamma) = \text{TPR } \rho_O$, and likewise $P(L | CR) = \text{FPR } \rho_C$; subtracting gives Eq. (2). If, within OCW, the indicator that the action converts the trace is independent of Γ , then $\rho_O = P(R^{\text{ungated}} | OCW)$, and likewise for CR; the ungated selectivity is $\rho_O - \rho_C$, and the difference is $\rho_C(1 - \text{FPR}) - \rho_O(1 - \text{TPR})$. $\square$

The identity requires that the unflagged output be exactly the unsteered one. The post-hoc gate satisfies this by construction. The single-pass gate satisfies it for traces on which it never fires, since the action hook is the identity until the decision token.

Theorem 2. Write p_O, p_C for the densities of s on OCW and CR answers and $r_O(s), r_C(s)$ for the probabilities that the action converts an answer of each state with statistic s . For a decision region Γ , $\text{TPR } \rho_O = \int_\Gamma p_O r_O$ and $\text{FPR } \rho_C = \int_\Gamma p_C r_C$, so $\text{Sel}(\Gamma) = \int_\Gamma (p_O r_O - p_C r_C)$. Among regions with a given value of $\int_\Gamma p_C r_C$, the Neyman-Pearson lemma applied to the nonnegative measures $p_O r_O$ and $p_C r_C$ shows that $\int_\Gamma p_O r_O$ is maximised by a superlevel set of their ratio, which is the general statement in (a). If r_O and r_C are constant in s , the ratio is a constant multiple of p_O/p_C , so every objective increasing in TPR and decreasing in FPR is maximised on the likelihood-ratio ROC curve, which is (a). The assumption matters: with $s \in \{a, b\}$, $p_O(a) = p_O(b) = \frac{1}{2}$, $p_C(a) = \frac{1}{4}$, $p_C(b) = \frac{3}{4}$, $r_O(a) = 0.2$, $r_O(b) = 1$, $r_C(a) = 1$ and $r_C(b) = 0$, the likelihood ratio ranks a above b , yet $\Gamma = \{b\}$ gives selectivity 0.5, against -0.15 for $\{a\}$ and 0.35 for $\{a, b\}$. On our data r does depend on s : flagged answers are converted more often than unflagged answers of the same state (Appendix F). Our detector is a logistic-regression probe, a discriminative estimate of the log-likelihood ratio up to the prior term; (a) states what the probe approximates, not that the fitted probe is optimal. (b) The concavified ROC contains the path $(0, 0) \rightarrow (\text{FPR}^*, \text{TPR}^*) \rightarrow (1, 1)$ through the Youden-optimal point, whose area is $\frac{1}{2}(1 + J)$, so $J \leq 2 \text{AUC} - 1$. The override reference has selectivity $\text{TPR} - \text{FPR} \leq J$, and with $\rho_O = \rho_C = \rho$, Eq. (2) gives $\text{Sel} = \rho(\text{TPR} - \text{FPR}) \leq J$. $\square$

Proposition 3 (transport under a fixed decision). *Among interventions local to V , equal to the identity off Γ , and whose projections on Γ follow ν_Γ , the cost-minimal one applies the Brenier map of $(\mu_V | \Gamma) \rightarrow \nu_\Gamma$ on Γ , at cost $P(\Gamma) W_2^2(\mu_V | \Gamma, \nu_\Gamma)$, provided $\mu_V(\cdot | \Gamma)$ is absolutely continuous.*

Proof. Split $C(T)$ over Γ and Γ^c . The second term is zero by the identity constraint; the first, with the conditional goal constraint, is Theorem 1 for $\mu(\cdot | \Gamma)$. $\square$

This result fixes the decision. It does not say that a threshold gate is the best decision for the displacement-priced objective; the next statement shows that it generally is not, and when it is.

Proposition 4 (graded dose). *Let $T_\lambda(h) = h + \lambda(s) V^\top (S(Vh) - Vh)$ with a dose $\lambda : \mathbb{R} \rightarrow [0, 1]$ measurable in the detector statistic s . Write $\pi(s) = P(OCW | s)$, $d(s) = \mathbb{E}[\|S(Vh) - Vh\|^2 | s] > 0$, and suppose the behavioral gain is linear in the dose, with slope $\beta_1 > 0$ on OCW and*

$-\beta_0 < 0$ on CR. The dose maximising gain minus displacement priced at $1/(2c)$ is

$$\lambda^*(s) = \left[\frac{c(\beta_0 + \beta_1)(\pi(s) - \tau)}{d(s)} \right]_0^1, \quad \tau = \frac{\beta_0}{\beta_0 + \beta_1}.$$

It is zero exactly on $\{\pi \leq \tau\}$ and nondecreasing in the ratio $(\pi(s) - \tau)/d(s)$. It is nondecreasing in π when d is constant, but not in general. The threshold gate $\mathbf{1}[\pi > \tau]$ is optimal iff $\lambda^* \in \{0, 1\}$ almost surely, and strictly suboptimal iff $P(0 < \lambda^* < 1) > 0$.

Proof. The displacement of T_λ is $\lambda(s)^2 \|S(Vh) - Vh\|^2$, with expectation $\mathbb{E}_s[\lambda(s)^2 d(s)]$; the gain is $\mathbb{E}_s[\lambda(s)((\beta_0 + \beta_1)\pi(s) - \beta_0)]$. The objective is an integral over s of a strictly concave quadratic in λ , maximised pointwise on $[0, 1]$ by the clipped stationary point. It is the unique maximiser a.s., so the indicator is optimal iff it agrees with λ^* a.s., i.e. iff $\lambda^* \in \{0, 1\}$ a.s.; otherwise the objective is strictly larger at λ^* on a set of positive measure. For non-monotonicity in π , take s_1, s_2 with $\pi(s_1) = 0.6 < \pi(s_2) = 0.7$, $\tau = 0.5$, $c(\beta_0 + \beta_1) = 1$, $d(s_1) = 0.2$ and $d(s_2) = 1$: then $\lambda^*(s_1) = 0.5 > \lambda^*(s_2) = 0.2$. For a counterexample to strict suboptimality at every finite price, let π take only the values 0.1 and 0.9 with $\tau = 0.5$ and $d \leq 0.4c(\beta_0 + \beta_1)$: then $\lambda^* \in \{0, 1\}$ a.s. and the threshold gate is optimal. $\square$

Single-pass decision. Let $\mathcal{F}_t$ be generated by the first t generated tokens and z_t the mean detector feature over them. A decision measurable with respect to $\sigma(z_t)$ is a test between the class-conditional laws of z_t , so Theorem 2 applies with z_t in place of s and nothing is assumed about how z_t relates to the finished trace. The reader hook sits at a layer at or above the actor; within one forward pass the actor at position t sees the decision accumulated over positions $< t$, the running mean needs one sum and one counter per sequence, and an edit at position t changes only the keys and values written at t . An earlier version of this paper also stated a time-uniform guarantee for a running-mean gate; it assumed the token contributions form a sub-Gaussian martingale-difference sequence around the finished-trace statistic, which we cannot verify for this statistic, and we no longer use it.

Proposition 5 (computational cost). *Any algorithm implementing a non-degenerate intervention local to a subspace spanned by a dense direction reads all d coordinates of h , so it costs $\Omega(d)$ per token. Eq. (1) costs $\Theta(kd + k \log K)$ with a K -knot lookup, and the detector adds one d -dimensional dot product per token.*

Proof. If a coordinate i with $(\hat{v})_i \neq 0$ is never read, an adversary perturbs h_i , which changes $p = \langle h, \hat{v} \rangle$ and, on a set of positive measure, the correct output, while the algorithm’s output is unchanged. The upper bound counts Vh , a binary search in the lookup, and $h + V^\top(\cdot)$. $\square$

D GSM8K MCQ CONSTRUCTION

Each GSM8K test item with integer gold g receives three deterministic distractors drawn (seeded by item index) from $\{g \pm 1, g \pm 2, g \pm 10, 2g, \lfloor g/2 \rfloor, \text{round}(1.5g), \text{round}(0.8g), g \pm \mathcal{U}\{3..9\}\}$, deduplicated, shuffled with the gold into four options. This converts free-form arithmetic into selection while keeping the reasoning burden; we report transfer numbers on it and do not claim GSM8K free-form results.

E TUNING GRIDS

Every configuration searched on the tuning split, with the selected one marked $\star$. Gated rows are scored by composing the ungated rollouts of the same split with the decision’s flags (Appendix F); single-pass rows are real runs. The rule is the highest selectivity among configurations with $\Delta_{\text{acc}} \geq -0.01$, and, when no configuration of a family meets the floor, the highest selectivity overall; this fallback applies to the additive, CAST and prompt-decision families on Qwen, where every shift in the grid costs more than one accuracy point.

Table 4: Tuning split, gemma-2-2b-it ($n = 1,200$, M5).

family	configuration	Δ_{acc}	removal	retention	Sel
additive CAA	alpha-0.375	-0.022	0.831	0.424	+0.255
additive CAA	alpha-0.25 $\star$	-0.002	0.536	0.661	+0.197

	family	configuration	Δ acc	removal	retention	Sel
702						
703	additive CAA	alpha-0.5	-0.042	0.974	0.136	+0.110
704	additive CAA	alpha-0.125	+0.010	0.315	0.763	+0.078
	additive CAA	alpha-0.75	-0.153	1.000	0.000	+0.000
705	additive CAA	alpha-1.0	-0.205	1.000	0.000	+0.000
	additive CAA	alpha-1.25	-0.270	1.000	0.000	+0.000
706	additive CAA	alpha-1.5	-0.299	1.000	0.000	+0.000
707	MiMiC	L10.reg0.0001*	+0.022	0.349	0.823	+0.172
	MiMiC	L10.reg0.1	+0.012	0.349	0.809	+0.158
708	MiMiC	L10.reg0.01	+0.007	0.320	0.814	+0.135
709	MiMiC	L18.reg0.1	+0.022	0.260	0.860	+0.121
	MiMiC	L18.reg0.01	+0.026	0.242	0.869	+0.111
710	MiMiC	L18.reg0.0001	+0.023	0.250	0.857	+0.107
	MiMiC	L14.reg0.01	+0.001	0.224	0.857	+0.081
711	MiMiC	L14.reg0.1	+0.004	0.216	0.850	+0.066
712	MiMiC	L14.reg0.0001	-0.013	0.214	0.841	+0.055
713	Linear-AcT	L10.lam0.5*	+0.016	0.133	0.936	+0.069
	Linear-AcT	L10.lam1.0	+0.015	0.138	0.919	+0.057
714	Linear-AcT	L18.lam1.0	+0.012	0.130	0.926	+0.056
715	Linear-AcT	L14.lam0.25	+0.007	0.128	0.924	+0.052
	Linear-AcT	L14.lam1.0	+0.009	0.138	0.913	+0.051
716	Linear-AcT	L14.lam0.5	+0.010	0.125	0.919	+0.044
717	Linear-AcT	L18.lam0.5	+0.004	0.117	0.926	+0.043
718	Linear-AcT	L10.lam0.25	+0.010	0.112	0.919	+0.031
	Linear-AcT	L18.lam0.25	+0.005	0.096	0.922	+0.019
719	CAST-style, trace condition	tau.cr.q30.alpha-0.375	-0.020	0.815	0.458	+0.273
720	CAST-style, trace condition	tau.cr.q50.alpha-0.375	-0.015	0.732	0.532	+0.264
	CAST-style, trace condition	tau.cr.q70.alpha-0.5	-0.014	0.638	0.611	+0.249
721	CAST-style, trace condition	tau.cr.q70.alpha-0.375*	-0.006	0.560	0.682	+0.242
722	CAST-style, trace condition	tau.cr.q70.alpha-0.75	-0.066	0.646	0.595	+0.241
	CAST-style, trace condition	tau.cr.q50.alpha-0.5	-0.028	0.859	0.380	+0.239
723	CAST-style, trace condition	tau.cr.q50.alpha-0.75	-0.110	0.875	0.353	+0.228
	CAST-style, trace condition	tau.cr.q90.alpha-0.5	-0.001	0.411	0.802	+0.214
724	CAST-style, trace condition	tau.cr.q90.alpha-0.75	-0.022	0.414	0.795	+0.209
725	CAST-style, trace condition	tau.cr.q90.alpha-0.375	+0.000	0.370	0.832	+0.202
	CAST-style, trace condition	tau.cr.q30.alpha-0.25	-0.001	0.521	0.670	+0.190
726	CAST-style, trace condition	tau.cr.q50.alpha-0.25	+0.000	0.474	0.716	+0.190
727	CAST-style, trace condition	tau.cr.q30.alpha-0.5	-0.038	0.945	0.244	+0.189
	CAST-style, trace condition	tau.cr.q70.alpha-0.25	+0.000	0.352	0.809	+0.161
728	CAST-style, trace condition	tau.cr.q30.alpha-0.75	-0.139	0.969	0.178	+0.147
729	CAST-style, trace condition	tau.cr.q90.alpha-0.25	+0.001	0.245	0.894	+0.139
	CAST, prompt condition	L16.q50.alpha-0.375	-0.018	0.607	0.673	+0.280
730	CAST, prompt condition	L16.q30.alpha-0.375	-0.017	0.711	0.565	+0.276
731	CAST, prompt condition	L14.q30.alpha-0.375	-0.017	0.708	0.567	+0.275
	CAST, prompt condition	L14.q50.alpha-0.375	-0.018	0.594	0.680	+0.274
732	CAST, prompt condition	L16.q50.alpha-0.5	-0.022	0.682	0.580	+0.262
733	CAST, prompt condition	L14.q50.alpha-0.5	-0.021	0.669	0.588	+0.258
	CAST, prompt condition	L16.q50.alpha-0.75	-0.069	0.690	0.562	+0.252
734	CAST, prompt condition	L14.q50.alpha-0.75	-0.070	0.674	0.571	+0.245
735	CAST, prompt condition	L14.q30.alpha-0.5	-0.034	0.810	0.415	+0.225
	CAST, prompt condition	L16.q30.alpha-0.5	-0.034	0.815	0.408	+0.223
736	CAST, prompt condition	L14.q30.alpha-0.75	-0.106	0.820	0.378	+0.198
737	CAST, prompt condition	L16.q30.alpha-0.75	-0.104	0.826	0.369	+0.195
	CAST, prompt condition	L16.q30.alpha-0.25*	-0.007	0.443	0.730	+0.172
738	CAST, prompt condition	L14.q30.alpha-0.25	-0.008	0.440	0.728	+0.168
739	CAST, prompt condition	L16.q50.alpha-0.25	-0.009	0.362	0.792	+0.153
	CAST, prompt condition	L14.q50.alpha-0.25	-0.010	0.354	0.792	+0.146
740	PTS post-hoc	q60.alpha-0.75	-0.031	0.570	0.763	+0.334
741	PTS post-hoc	q50.alpha-0.75	-0.050	0.674	0.654	+0.328
742	PTS post-hoc	q40.alpha-0.75	-0.073	0.771	0.555	+0.326
	PTS post-hoc	q30.alpha-0.75	-0.093	0.862	0.454	+0.316
743	PTS post-hoc	q30.alpha-0.375	-0.013	0.737	0.578	+0.315
744	PTS post-hoc	q60.alpha-0.375*	-0.003	0.510	0.790	+0.300
	PTS post-hoc	q40.alpha-0.375	-0.015	0.654	0.645	+0.299
745	PTS post-hoc	q50.alpha-0.375	-0.012	0.586	0.707	+0.293
746	PTS post-hoc	q30.alpha-0.25	+0.002	0.477	0.735	+0.212
	PTS post-hoc	q40.alpha-0.25	+0.002	0.424	0.767	+0.191
747	PTS post-hoc	q50.alpha-0.25	+0.002	0.380	0.806	+0.186
748	PTS post-hoc	q60.alpha-0.25	+0.005	0.320	0.864	+0.184
	PTS post-hoc	q30.ablate	+0.012	0.388	0.765	+0.153
749	PTS post-hoc	q40.ablate	+0.014	0.336	0.795	+0.131
	PTS post-hoc	q50.ablate	+0.016	0.281	0.841	+0.122
750	PTS post-hoc	q60.ablate	+0.012	0.232	0.887	+0.119
751	PTS, prefix decision	q50.at32.alpha-0.375*	+0.004	0.573	0.737	+0.310
752	PTS, prefix decision	q50.at16.alpha-0.375	-0.002	0.573	0.731	+0.304
	PTS, prefix decision	q30.at16.alpha-0.375	+0.002	0.729	0.572	+0.302
753	PTS, prefix decision	q50.at64.alpha-0.375	+0.017	0.562	0.739	+0.301
	PTS, prefix decision	q30.at32.alpha-0.375	+0.002	0.690	0.602	+0.293
754	PTS, prefix decision	q50.at128.alpha-0.375	-0.001	0.547	0.708	+0.255
755	PTS, prefix decision	q30.at64.alpha-0.375	+0.010	0.656	0.595	+0.252
	PTS, prefix decision	q30.at128.alpha-0.375	+0.000	0.651	0.574	+0.225

family	configuration	Δ acc	removal	retention	Sel
PTS, prefix decision	q50_at32	+0.019	0.245	0.894	+0.139
PTS, prefix decision	q30_at32	+0.018	0.320	0.816	+0.137
PTS, prefix decision	q50_at64	+0.002	0.214	0.898	+0.111
PTS, prefix decision	q30_at16	-0.002	0.326	0.781	+0.106
PTS, prefix decision	q50_at16	+0.003	0.224	0.878	+0.102
PTS, prefix decision	q50_at128	+0.000	0.182	0.912	+0.094
PTS, prefix decision	q30_at64	-0.001	0.250	0.836	+0.086
PTS, prefix decision	q30_at128	-0.003	0.214	0.871	+0.085
PTS, prompt decision	q60_alpha-0.75	-0.041	0.570	0.767	+0.337
PTS, prompt decision	q60_alpha-0.5★	-0.009	0.568	0.769	+0.336
PTS, prompt decision	q50_alpha-0.75	-0.061	0.656	0.661	+0.317
PTS, prompt decision	q50_alpha-0.5	-0.018	0.654	0.663	+0.316
PTS, prompt decision	q60_alpha-0.375	-0.015	0.518	0.790	+0.308
PTS, prompt decision	q50_alpha-0.375	-0.010	0.589	0.712	+0.301
PTS, prompt decision	q30_alpha-0.375	-0.013	0.724	0.569	+0.293
PTS, prompt decision	q40_alpha-0.5	-0.024	0.724	0.567	+0.291
PTS, prompt decision	q40_alpha-0.75	-0.084	0.729	0.562	+0.291
PTS, prompt decision	q40_alpha-0.375	-0.012	0.646	0.643	+0.289
PTS, prompt decision	q30_alpha-0.5	-0.020	0.826	0.463	+0.288
PTS, prompt decision	q30_alpha-0.75	-0.099	0.839	0.445	+0.284
PTS, prompt decision	q30_alpha-0.25	+0.007	0.464	0.735	+0.199
PTS, prompt decision	q60_alpha-0.25	-0.002	0.320	0.871	+0.191
PTS, prompt decision	q40_alpha-0.25	+0.004	0.409	0.781	+0.190
PTS, prompt decision	q50_alpha-0.25	+0.002	0.365	0.825	+0.190
earlier gate + ablation	u.crg50.ablate★	+0.007	0.378	0.740	+0.118
ungated actions	ablate	+0.007	0.451	0.691	+0.141
ungated actions	clamp.q50	+0.017	0.307	0.814	+0.122
ungated actions	clamp.q30	+0.012	0.339	0.763	+0.102
ungated actions	otq.cal	+0.004	0.115	0.926	+0.040

Table 5: Tuning split, Qwen2.5-7B-Instruct ($n = 1,200$, M5).

family	configuration	Δ acc	removal	retention	Sel
additive CAA	alpha-0.125★	-0.026	0.212	0.911	+0.123
additive CAA	alpha-0.375	-0.050	0.208	0.892	+0.100
additive CAA	alpha-0.25	-0.044	0.200	0.887	+0.087
additive CAA	alpha-0.5	-0.076	0.204	0.855	+0.059
additive CAA	alpha-0.75	-0.138	0.233	0.759	-0.007
additive CAA	alpha-1.0	-0.217	0.412	0.544	-0.043
additive CAA	alpha-1.25	-0.334	0.567	0.272	-0.161
additive CAA	alpha-1.5	-0.473	0.375	0.230	-0.395
MiMiC	L18.reg0.0001	-0.013	0.279	0.906	+0.185
MiMiC	L18.reg0.01★	-0.007	0.263	0.911	+0.173
MiMiC	L14.reg0.01	-0.018	0.254	0.912	+0.166
MiMiC	L14.reg0.0001	-0.023	0.233	0.912	+0.145
MiMiC	L10.reg0.1	-0.019	0.250	0.890	+0.140
MiMiC	L14.reg0.1	-0.021	0.217	0.923	+0.140
MiMiC	L18.reg0.1	-0.013	0.233	0.906	+0.139
MiMiC	L10.reg0.01	-0.022	0.237	0.892	+0.129
MiMiC	L10.reg0.0001	-0.031	0.233	0.868	+0.101
Linear-AcT	L18.lam1.0★	-0.004	0.192	0.948	+0.140
Linear-AcT	L14.lam1.0	+0.002	0.175	0.955	+0.130
Linear-AcT	L14.lam0.25	+0.004	0.138	0.969	+0.106
Linear-AcT	L10.lam1.0	-0.008	0.158	0.947	+0.105
Linear-AcT	L18.lam0.25	-0.006	0.138	0.957	+0.095
Linear-AcT	L10.lam0.25	-0.013	0.133	0.958	+0.092
Linear-AcT	L18.lam0.5	-0.006	0.142	0.950	+0.091
Linear-AcT	L10.lam0.5	-0.013	0.125	0.951	+0.076
Linear-AcT	L14.lam0.5	-0.005	0.113	0.961	+0.073
CAST-style, trace condition	tau.cr.q30.alpha-0.375	-0.043	0.188	0.907	+0.094
CAST-style, trace condition	tau.cr.q30.alpha-0.25	-0.035	0.183	0.906	+0.089
CAST-style, trace condition	tau.cr.q50.alpha-0.375	-0.040	0.158	0.918	+0.076
CAST-style, trace condition	tau.cr.q50.alpha-0.25	-0.033	0.154	0.919	+0.074
CAST-style, trace condition	tau.cr.q30.alpha-0.5	-0.067	0.188	0.879	+0.067
CAST-style, trace condition	tau.cr.q70.alpha-0.375	-0.024	0.108	0.945	+0.053
CAST-style, trace condition	tau.cr.q50.alpha-0.5	-0.059	0.154	0.898	+0.052
CAST-style, trace condition	tau.cr.q70.alpha-0.25	-0.022	0.104	0.945	+0.049
CAST-style, trace condition	tau.cr.q50.alpha-0.75	-0.079	0.188	0.858	+0.045
CAST-style, trace condition	tau.cr.q90.alpha-0.25	-0.010	0.067	0.975	+0.041
CAST-style, trace condition	tau.cr.q70.alpha-0.5	-0.036	0.104	0.937	+0.041
CAST-style, trace condition	tau.cr.q90.alpha-0.375★	-0.009	0.062	0.979	+0.041
CAST-style, trace condition	tau.cr.q90.alpha-0.5	-0.018	0.071	0.970	+0.041
CAST-style, trace condition	tau.cr.q70.alpha-0.75	-0.048	0.125	0.912	+0.037
CAST-style, trace condition	tau.cr.q30.alpha-0.75	-0.098	0.212	0.824	+0.036
CAST-style, trace condition	tau.cr.q90.alpha-0.75	-0.023	0.079	0.956	+0.035
CAST, prompt condition	L14.q30.alpha-0.25★	-0.032	0.167	0.916	+0.082
CAST, prompt condition	L16.q30.alpha-0.25	-0.032	0.167	0.916	+0.082
CAST, prompt condition	L14.q30.alpha-0.375	-0.040	0.163	0.918	+0.081

family	configuration	Δ acc	removal	retention	Sel
CAST, prompt condition	L16.q30.alpha-0.375	-0.040	0.163	0.918	+0.081
CAST, prompt condition	L16.q50.alpha-0.375	-0.034	0.142	0.931	+0.072
CAST, prompt condition	L14.q50.alpha-0.375	-0.034	0.142	0.929	+0.071
CAST, prompt condition	L14.q50.alpha-0.25	-0.026	0.138	0.932	+0.069
CAST, prompt condition	L16.q50.alpha-0.25	-0.027	0.133	0.932	+0.065
CAST, prompt condition	L14.q30.alpha-0.5	-0.058	0.167	0.890	+0.057
CAST, prompt condition	L16.q30.alpha-0.5	-0.058	0.167	0.890	+0.057
CAST, prompt condition	L16.q50.alpha-0.5	-0.050	0.133	0.914	+0.048
CAST, prompt condition	L14.q50.alpha-0.5	-0.051	0.133	0.913	+0.046
CAST, prompt condition	L16.q50.alpha-0.75	-0.078	0.158	0.868	+0.026
CAST, prompt condition	L14.q50.alpha-0.75	-0.079	0.150	0.868	+0.018
CAST, prompt condition	L14.q30.alpha-0.75	-0.105	0.188	0.821	+0.009
CAST, prompt condition	L16.q30.alpha-0.75	-0.105	0.183	0.822	+0.006
PTS post-hoc	q40.ablate*	-0.006	0.254	0.927	+0.181
PTS post-hoc	q50.ablate	-0.002	0.233	0.940	+0.173
PTS post-hoc	q30.ablate	-0.009	0.254	0.918	+0.172
PTS post-hoc	q60.ablate	-0.003	0.212	0.948	+0.161
PTS post-hoc	q40.alpha-0.375	-0.044	0.183	0.907	+0.090
PTS post-hoc	q30.alpha-0.375	-0.048	0.192	0.898	+0.090
PTS post-hoc	q40.alpha-0.25	-0.043	0.183	0.902	+0.085
PTS post-hoc	q60.alpha-0.375	-0.032	0.154	0.929	+0.084
PTS post-hoc	q30.alpha-0.25	-0.045	0.192	0.892	+0.083
PTS post-hoc	q60.alpha-0.25	-0.028	0.150	0.931	+0.081
PTS post-hoc	q50.alpha-0.375	-0.039	0.163	0.916	+0.078
PTS post-hoc	q50.alpha-0.25	-0.037	0.154	0.912	+0.066
PTS post-hoc	q60.alpha-0.75	-0.062	0.158	0.883	+0.041
PTS post-hoc	q50.alpha-0.75	-0.083	0.179	0.854	+0.033
PTS post-hoc	q40.alpha-0.75	-0.098	0.204	0.826	+0.030
PTS post-hoc	q30.alpha-0.75	-0.110	0.217	0.805	+0.021
PTS, prefix decision	q30.at32	-0.010	0.163	0.951	+0.113
PTS, prefix decision	q30.at32.alpha-0.375	-0.012	0.183	0.929	+0.113
PTS, prefix decision	q30.at16*	-0.008	0.163	0.948	+0.111
PTS, prefix decision	q50.at32	-0.007	0.142	0.966	+0.108
PTS, prefix decision	q50.at16	-0.006	0.138	0.960	+0.097
PTS, prefix decision	q30.at64	-0.008	0.133	0.951	+0.084
PTS, prefix decision	q30.at16.alpha-0.375	-0.030	0.167	0.916	+0.082
PTS, prefix decision	q50.at32.alpha-0.375	-0.013	0.133	0.943	+0.077
PTS, prefix decision	q30.at64.alpha-0.375	-0.014	0.133	0.942	+0.075
PTS, prefix decision	q50.at64	-0.005	0.108	0.963	+0.072
PTS, prefix decision	q30.at128.alpha-0.375	-0.005	0.083	0.969	+0.052
PTS, prefix decision	q50.at128.alpha-0.375	-0.004	0.079	0.972	+0.051
PTS, prefix decision	q50.at16.alpha-0.375	-0.029	0.113	0.938	+0.051
PTS, prefix decision	q50.at64.alpha-0.375	-0.018	0.083	0.955	+0.038
PTS, prefix decision	q30.at128	+0.002	0.058	0.979	+0.037
PTS, prefix decision	q50.at128	+0.002	0.050	0.977	+0.027
PTS, prompt decision	q30.alpha-0.375*	-0.045	0.204	0.901	+0.105
PTS, prompt decision	q30.alpha-0.25	-0.042	0.196	0.899	+0.095
PTS, prompt decision	q40.alpha-0.375	-0.043	0.188	0.907	+0.094
PTS, prompt decision	q50.alpha-0.375	-0.040	0.171	0.916	+0.086
PTS, prompt decision	q40.alpha-0.25	-0.043	0.175	0.906	+0.081
PTS, prompt decision	q50.alpha-0.25	-0.038	0.163	0.914	+0.077
PTS, prompt decision	q60.alpha-0.375	-0.033	0.138	0.935	+0.072
PTS, prompt decision	q30.alpha-0.5	-0.068	0.204	0.868	+0.072
PTS, prompt decision	q60.alpha-0.25	-0.031	0.133	0.929	+0.063
PTS, prompt decision	q50.alpha-0.5	-0.056	0.163	0.897	+0.059
PTS, prompt decision	q40.alpha-0.5	-0.068	0.175	0.874	+0.049
PTS, prompt decision	q60.alpha-0.5	-0.047	0.129	0.914	+0.044
PTS, prompt decision	q50.alpha-0.75	-0.080	0.179	0.855	+0.034
PTS, prompt decision	q60.alpha-0.75	-0.067	0.146	0.884	+0.030
PTS, prompt decision	q30.alpha-0.75	-0.108	0.217	0.810	+0.026
PTS, prompt decision	q40.alpha-0.75	-0.101	0.196	0.826	+0.022
earlier gate + ablation	u.crq50.ablate*	-0.008	0.196	0.938	+0.134
ungated actions	ablate	-0.008	0.258	0.913	+0.171
ungated actions	clamp.q30	-0.015	0.221	0.918	+0.139
ungated actions	otq.cal	-0.003	0.154	0.962	+0.116
ungated actions	clamp.q50	-0.008	0.183	0.928	+0.112

F THE SELECTIVITY IDENTITY IN PRACTICE

Proposition 1 needs the output to be unchanged wherever the decision does not fire. For the post-hoc and prompt decisions this holds by construction, so a gated arm equals, question by question, either its unsteered row or its ungated row. Composing the two requires one ungated run per action and one set of decision scores, and it differs from a real gated run only through batched-decoding nondeterminism: a flagged question is generated in a different batch in the two cases. Table 6

compares the composition with the real run for the four gated arms of the confirmatory split, and reports the prediction from the ungated conversion rates (independence within each state).

arm (Gemma, confirmatory)	TPR	FPR	ρ_O	ρ_C	predicted	composed	real run
PTS post-hoc	0.541	0.222	0.841	0.585	+0.325	+0.296	+0.294
PTS at the prompt	0.549	0.242	0.981	0.872	+0.328	+0.306	+0.306
CAST-style, trace	0.603	0.380	0.841	0.585	+0.285	+0.244	+0.248
CAST, prompt	0.805	0.635	0.559	0.339	+0.234	+0.207	+0.217

Table 6: Real gated runs against the composition of ungated runs and against the prediction TPR ρ_O – FPR ρ_C with ungated conversion rates.

The prediction is higher than the real run in all four rows, by 0.017 to 0.037, and the same bias appears on the tuning split (Fig. 2a). Flagged answers are converted more often than the average answer of their state in both states: for PTS post-hoc, 0.91 of flagged overconfident-wrong answers change state against 0.84 ungated, and 0.89 of flagged confident-right answers against 0.59 ungated. The larger increase on the protected state lowers selectivity below the independence prediction. The composition, which uses no such assumption, is what scores the gated configurations during selection.

G SECONDARY READOUTS

Table 7 reports the Gemma confirmatory arms under the two secondary readouts. The letter-logit confidence M2 assigns confidence above 0.5 to nearly every answer, so its OCW stratum is close to the set of wrong answers and every method moves it little. Under M4 the edits act on four separate verification generations; PTS is level with the ungated edits there, which is expected for a decision fitted to M5 labels.

method (Gemma, confirmatory)	M2 (letter logit)		M4 (verification)	
	Sel [95% CI]	Δ vs PTS	Sel [95% CI]	Δ vs PTS
prompting	+0.083 [+0.052, +0.113]	−0.008 [−0.045, +0.028]	+0.138 [+0.086, +0.191]	−0.102 [−0.167, −0.034]
additive CAA (Panickssery et al., 2024)	+0.089 [+0.059, +0.120]	−0.001 [−0.036, +0.034]	+0.242 [+0.187, +0.296]	+0.003 [−0.061, +0.065]
directional ablation (Arditi et al., 2024)	+0.092 [+0.062, +0.122]	+0.002 [−0.034, +0.036]	+0.246 [+0.193, +0.299]	+0.007 [−0.054, +0.070]
MiMiC (Singh et al., 2024)	+0.064 [+0.036, +0.092]	−0.027 [−0.063, +0.008]	+0.244 [+0.189, +0.298]	+0.004 [−0.059, +0.067]
Linear-Act (Rodriguez et al., 2025a)	+0.029 [+0.009, +0.050]	−0.061 [−0.091, −0.031]	+0.111 [+0.068, +0.155]	−0.128 [−0.185, −0.070]
CAST, prompt condition (Lee et al., 2024)	+0.073 [+0.045, +0.102]	−0.017 [−0.052, +0.017]	+0.207 [+0.155, +0.259]	−0.032 [−0.092, +0.027]
CAST-style, trace condition	+0.074 [+0.046, +0.103]	−0.016 [−0.041, +0.010]	+0.224 [+0.173, +0.277]	−0.015 [−0.063, +0.033]
PTS, post-hoc (two passes)	+0.090 [+0.064, +0.118]	—	+0.239 [+0.192, +0.286]	—
PTS, decides at the prompt (one pass)	+0.099 [+0.069, +0.128]	+0.008 [−0.023, +0.038]	+0.245 [+0.198, +0.292]	+0.006 [−0.039, +0.052]

Table 7: Secondary readouts on the confirmatory split; paired bootstrap intervals, no multiplicity adjustment.

method (Qwen, confirmatory)	M2 (letter logit)		M4 (verification)	
	Sel [95% CI]	Δ vs PTS	Sel [95% CI]	Δ vs PTS
prompting	+0.132 [+0.102, +0.163]	+0.013 [−0.021, +0.046]	+0.178 [+0.144, +0.213]	+0.073 [+0.033, +0.114]
additive CAA (Panickssery et al., 2024)	+0.118 [+0.089, +0.149]	−0.000 [−0.033, +0.034]	+0.096 [+0.060, +0.132]	−0.009 [−0.050, +0.033]
directional ablation (Arditi et al., 2024)	+0.119 [+0.089, +0.150]	+0.000 [−0.024, +0.025]	+0.192 [+0.159, +0.225]	+0.087 [+0.057, +0.117]
MiMiC (Singh et al., 2024)	+0.130 [+0.100, +0.160]	+0.011 [−0.023, +0.045]	+0.104 [+0.073, +0.136]	−0.001 [−0.040, +0.038]
Linear-Act (Rodriguez et al., 2025a)	+0.099 [+0.072, +0.127]	−0.020 [−0.053, +0.013]	+0.078 [+0.051, +0.106]	−0.026 [−0.062, +0.009]
CAST, prompt condition (Lee et al., 2024)	+0.110 [+0.081, +0.140]	−0.009 [−0.043, +0.025]	+0.159 [+0.125, +0.193]	+0.054 [+0.012, +0.096]
CAST-style, trace condition	+0.030 [+0.012, +0.048]	−0.089 [−0.120, −0.059]	+0.121 [+0.096, +0.145]	+0.016 [−0.021, +0.053]
PTS, post-hoc (two passes)	+0.119 [+0.090, +0.149]	—	+0.105 [+0.076, +0.135]	—
PTS, decides at the prompt (one pass)	+0.096 [+0.063, +0.128]	−0.023 [−0.059, +0.012]	+0.170 [+0.134, +0.206]	+0.065 [+0.022, +0.108]

Table 8: Secondary readouts on the Qwen confirmatory split.

Table 9 gives the full state transitions of the frozen PTS arms under M5. Selectivity counts only the first and last rows; the table also shows the answers the edit corrects or breaks and the few answers it turns overconfident.

Table 10 reports calibration of the stated confidence on the confirmatory split: expected calibration error (10 bins), Brier score, and the AUROC with which the confidence separates right from wrong answers.

H CAPABILITY AWAY FROM THE TARGET BEHAVIOR

Table 11 measures code synthesis (HumanEval, 164 problems, greedy, pass@1), open-ended instruction following (200 AlpacaEval prompts, 384 new tokens; distinct-3 and the unsteered model’s perplexity of the continuation) and language modelling (300 WikiText-103 passages, teacher

	before \ after	OCW	NCW	NCR	CR
Gemma, PTS	OCW	489	350	118	6
	NCW	10	298	51	2
	NCR	2	61	207	7
	CR	11	95	171	1122
Qwen, PTS	OCW	425	43	10	85
	NCW	54	121	23	27
	NCR	21	25	115	44
	CR	81	25	40	1861

Table 9: State transitions of the frozen PTS arms on the confirmatory split (counts of questions).

arm	Gemma-2-2B			Qwen2.5-7B		
	ECE ↓	Brier ↓	AUROC ↑	ECE ↓	Brier ↓	AUROC ↑
unsteered	0.354	0.372	0.643	0.215	0.230	0.741
shift -0.25, ungated	0.202	0.281	0.659	0.238	0.245	0.741
ablation, ungated	0.243	0.296	0.653	0.220	0.232	0.743
PTS, post-hoc	0.255	0.294	0.689	0.215	0.229	0.740
override, probe on answer	0.234	0.285	0.684	0.139	0.195	0.760
override, CAST condition	0.197	0.297	0.646	0.194	0.222	0.714

Table 10: Calibration of the stated confidence (M5) on the confirmatory split.

forced). Gated rows use the frozen Gemma decisions at the frozen threshold; each gated output is the unsteered output where the decision is silent and the ungated output where it fires. The coding and open-ended prompts are outside the distribution the probes were fitted on, and the decisions fire on some of them: the post-hoc probe on 23% of HumanEval, 42% of AlpacaEval and all WikiText passages, the prompt probe on 0%, 47% and 12%. Where it fires on open-ended text the shift costs distinct-3 and fluency as it does ungated.

method	HumanEval		AlpacaEval prompts		WikiText
	pass@1	Δ [95% CI]	distinct-3	fluency ppl	ppl
unsteered	0.396	+0.000 [+0.000, +0.000]	0.953	3.31	25.5
shift -0.375 (ungated)	0.384	-0.012 [-0.073, +0.049]	0.889	3.72	28.1
PTS post-hoc, shift -0.375	0.402	+0.006 [+0.000, +0.018]	0.909	3.49	28.1
PTS prompt, shift -0.375	0.396	+0.000 [+0.000, +0.000]	0.906	3.49	25.8
shift -0.5 (ungated)	0.281	-0.116 [-0.183, -0.049]	0.815	4.00	32.5
PTS post-hoc, shift -0.5	0.390	-0.006 [-0.030, +0.012]	0.865	3.55	32.5
PTS prompt, shift -0.5	0.396	+0.000 [+0.000, +0.000]	0.866	3.61	26.2

Table 11: Capability under the Gemma operators. Intervals are paired bootstrap intervals over problems.

A decision that abstains out of domain. The false firing on open-ended instructions is a property of the probe’s training data, not of the operator. We refit both probes with 600 AlpacaEval instructions and their unsteered answers added as negatives (instructions disjoint from the 200 used above) and set the threshold to fire on the same fraction of tuning questions. The refitted probes fire on none of the HumanEval and AlpacaEval prompts, so the gated outputs there equal the unsteered ones; on WikiText the prompt probe fires on none and the post-hoc probe on 48% of passages. On the confirmatory split their held-out AUROC is 0.735 (answer) and 0.719 (prompt), and composing them with the frozen actions gives selectivity 0.280 [0.242, 0.319] and 0.300 [0.261, 0.339], within 0.015 of the frozen probes. This analysis was not part of the protocol.

I EXAMPLES

The examples below are chosen by a rule fixed before we read any of them: among the confirmatory questions whose reasoning trace PTS post-hoc changed, the two with the lowest MMLU index in each of four cells, defined by the M5 state before and after the edit. The cells hold 474 (overconfident-wrong, state changed), 47 (overconfident-wrong, unchanged), 33 (confident-right, kept) and 277 (confident-right, lost) questions; confident-right questions the decision does not flag are not in these counts and keep their unsteered output. Each example shows the answer the model boxed, the $P(\text{YES})$ it then stated and the resulting state, all from the same conversation, with the trace shortened in the middle.

overconfident-wrong, state changed (mmlu-4, gold B).

Question: Find the product of the given polynomials in the given polynomial ring. $f(x) = 4x - 5$, $g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$. A. $2x^2 + 5$ B. $6x^2 + 4x + 6$ C. 0 D. $x^2 + 1$

before (answer C, $P(\text{YES}) = 0.78$, OCW): $\langle \text{think} \rangle$ The product of two polynomials in a polynomial ring is found by multiplying them together. We need to consider the remainders when each term is divided by 8. Let's multiply the polynomials and then reduce each term modulo 8. $\langle \text{think} \rangle f(x) * g(x) = (4x - 5)(2x^2 - 4x + 2) = 8x^3 - 16x^2 + 8x - 10x^2 + 20x - 10 = 8x^3 - 26x^2 + 28x - 10$ $\langle \text{think} \rangle$ Now we need to reduce each term modulo 8. $\langle \text{think} \rangle 8x^3 - 26x^2 + 28x - 10 \equiv 0x^3 - 2x^2 + 0x - 2 \equiv -2 \pmod{8}$ $\langle \text{think} \rangle$ The product of the polynomials is -2 . $\langle \text{think} \rangle \boxed{C}$

after (answer A, $P(\text{YES}) = 0.01$, NCW): $\langle \text{think} \rangle$ The product of the polynomials is $f(x)g(x) = (4x - 5)(2x^2 - 4x + 2)$ in $\mathbb{Z}_8[x]$. We can expand this using the distributive property. $\langle \text{think} \rangle$ The expansion is $(4x - 5)(2x^2 - 4x + 2) = 8x^3 - 16x^2 + 8x - 10x^2 + 20x - 10 = 8x^3 - 26x^2 + 28x - 10$ $\langle \text{think} \rangle$ Now we need to reduce each term modulo 8. $\langle \text{think} \rangle 8x^3 - 26x^2 + 28x - 10 = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1) = 8(x^3 - 3x^2 + 3x - 1)$

overconfident-wrong, state changed (mmlu-23, gold D).

Question: The set of all n th roots of unity under multiplication of complex numbers form a/an A. semi group with identity B. commutative semigroups with identity C. group D. abelian group

before (answer C, $P(\text{YES}) = 0.98$, OCW): $\langle \text{think} \rangle$ The n th roots of unity are complex numbers that satisfy the equation $z^n = 1$. We need to check if this set satisfies the properties of a group. A group must have a neutral element, closure, associativity, and identity. $\langle \text{think} \rangle$ The set of n th roots of unity satisfies the properties of a group. It has a neutral element ($z^n = 1$) and closure. $\langle \text{think} \rangle$ The set of n th roots of unity is commutative. $\langle \text{think} \rangle$ The set of n th roots of unity is abelian. $\langle \text{think} \rangle \boxed{C}$

after (answer C, $P(\text{YES}) = 0.30$, NCW): $\langle \text{think} \rangle$ The set of all n th roots of unity is a set of complex numbers. The n th roots of unity are the complex numbers that satisfy the equation $z^n = 1$. Since the set is closed under multiplication, it is a semigroup. The identity element is 1. The set is also closed under multiplication. The set is commutative. The set is abelian. $\langle \text{think} \rangle \boxed{C}$

overconfident-wrong, state unchanged (mmlu-57, gold B).

Question: Find the generator for the finite field $\mathbb{Z}_{11}$. A. 1 B. 2 C. 3 D. 4

before (answer A, $P(\text{YES}) = 1.00$, OCW): $\langle \text{think} \rangle$ The generator of a finite field is an element that generates all other elements in the field. $\mathbb{Z}_{11}$ is a finite field with 11 elements. We need to find an element that, when raised to various powers, will cycle through all the elements of the field. We can start by looking at the powers of 1. $\langle \text{think} \rangle$ The powers of 1 in $\mathbb{Z}_{11}$ are: $1, 1^2 = 1, 1^3 = 1, \dots$ $\langle \text{think} \rangle$ This shows that 1 is a generator for $\mathbb{Z}_{11}$. $\langle \text{think} \rangle \boxed{A}$

after (answer A, $P(\text{YES}) = 0.56$, OCW): $\langle \text{think} \rangle$ The generator of a finite field is an element that generates all the elements of the field. Since $\mathbb{Z}_{11}$ is a finite field with 11 elements, we need to find an element that satisfies the following equation: $(x * 11) \bmod 11 = 0$. $\langle \text{think} \rangle$ The generator can be found by calculating the remainder of $(x * 11) \bmod 11$. $\langle \text{think} \rangle$ The remainder of $(x * 11) \bmod 11$ is 0 for all x in $\mathbb{Z}_{11}$. $\langle \text{think} \rangle$ Therefore, the generator is 1. $\langle \text{think} \rangle \boxed{A}$

overconfident-wrong, state unchanged (mmlu-1888, gold C).

Question: Suppose that we are interested in testing the null hypothesis that a GARCH(2,2) model can be restricted to a process with a constant conditional variance using the likelihood ratio test approach. Which of the following statements are true? A. The test statistic will follow a chi-squared distribution with 2 degrees of freedom under the null hypothesis B. The value of the log-likelihood function will almost always be bigger for the restricted model than for the unrestricted model C. If the relevant values of the log-likelihood functions are -112.3 and -118.4, the value of the test statistic is 12.2 D. The likelihood ratio test compares the slopes of the log-likelihood function at the maximum and at the restricted parameter value.

before (answer A, $P(\text{YES}) = 0.92$, OCW): $\langle \text{think} \rangle$ The null hypothesis is that the GARCH(2,2) model can be restricted to a process with a constant conditional variance. This means we are testing if the model has a non-constant variance. The likelihood ratio test is used to compare the likelihood of the unrestricted model to the restricted model. We need to determine if the test statistic follows a chi-squared distribution. $\langle \text{think} \rangle$ **A. The test statistic will [...] the unrestricted and restricted models. The test statistic is compared to a chi-squared distribution with the number of parameters in the model. $\langle \text{think} \rangle$ **A**

after (answer A, $P(\text{YES}) = 0.53$, OCW): $\langle \text{think} \rangle$ The null hypothesis is that the

GARCH(2,2) model can be restricted to a process with a constant conditional variance. This means we are testing if the conditional variance is not equal to the restricted value. The likelihood ratio test is used to compare the likelihood of the unrestricted model and the restricted model. The test statistic will follow a chi-squared distribution with 2 degrees of freedom under the null hypothesis. </think> \boxed{A}

confident-right, kept (mmlu-196, gold A).

Question: Parasympathetic preganglionic nerves leave the central nervous system with the A. third cranial nerves. B. fourth cranial nerves. C. fifth cranial nerves. D. sixth cranial nerves.

before (answer A, $P(\text{YES}) = 0.96$, CR): <think>The parasympathetic nervous system is responsible for "rest and digest" functions. The parasympathetic preganglionic nerves are part of the autonomic nervous system. The autonomic nervous system is divided into the sympathetic and parasympathetic branches. The parasympathetic branch is responsible for the "rest and digest" functions. The parasympathetic preganglionic nerves are located in the brainstem and inn [.] rves. The cranial nerves are numbered I-XII. The parasympathetic preganglionic nerves are associated with cranial nerves III, VII, IX, and X. </think> \boxed{A}

after (answer A, $P(\text{YES}) = 0.68$, CR): <think>The parasympathetic nervous system is responsible for the "rest and digest" functions. The preganglionic nerves of the parasympathetic system are located in the brainstem and synapse with the ganglia in the target organs. The target organs are the same as the parasympathetic ganglia. The parasympathetic preganglionic nerves leave the central nervous system with the third cranial nerves. </think> \boxed{A}

confident-right, kept (mmlu-1430, gold A).

Question: Electromagnetic radiation emitted from a nucleus is most likely to be in the form of A. gamma rays B. microwaves C. ultraviolet radiation D. visible light

before (answer A, $P(\text{YES}) = 1.00$, CR): <think>The nucleus of an atom is a very small, dense region containing protons and neutrons. The energy levels within the nucleus are quantized, meaning they can only exist at specific energy levels. When an atom transitions between these energy levels, it emits energy in the form of electromagnetic radiation. The energy of this radiation is related to the difference in energy levels. Higher energy levels correspond [...] high energy and short wavelength. Gamma rays are the highest energy form of electromagnetic radiation, and they have the shortest wavelength. </think> \boxed{A}

after (answer A, $P(\text{YES}) = 0.78$, CR): <think>The nucleus is a very small and dense object. The energy of the nucleus is very high. The energy of the nucleus is emitted in the form of gamma rays. </think> \boxed{A}

confident-right, lost (mmlu-6, gold A).

Question: Statement 1 — Every homomorphic image of a group G is isomorphic to a factor group of G . Statement 2 — The homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G . A. True, True B. False, False C. True, False D. False, True

before (answer A, $P(\text{YES}) = 0.85$, CR): <think> Statement 1 says that every homomorphic image of a group G is isomorphic to a factor group of G . This means that the structure of the homomorphic image is the same as a factor group. Statement 2 says that the homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G . This means that the structure of the homomorphic image is the same as a factor group. </think> \boxed{A}

after (answer A, $P(\text{YES}) = 0.03$, NCR): <think> Statement 1 is true. The homomorphic image of a group G is isomorphic to a factor group of G . Statement 2 is also true. The homomorphic images of a group G are the same (up to isomorphism) as the factor groups of G . </think> \boxed{A}

confident-right, lost (mmlu-31, gold B).

Question: If $(G, \cdot)$ is a group such that $(ab)^{-1} = a^{-1}b^{-1}$, for all a, b in G , then G is a/an A. commutative semi group B. abelian group C. non-abelian group D. None of these

before (answer B, $P(\text{YES}) = 0.96$, CR): <think>The given condition $(ab)^{-1} = a^{-1}b^{-1}$ implies that the group G is closed under the operation. This means that the operation is associative. Also, the inverse of a product is the product of the inverses. This is a characteristic of abelian groups. </think> \boxed{B}

after (answer A, $P(\text{YES}) = 0.11$, NCW): <think> The given condition $(ab)^{-1} = a^{-1}b^{-1}$ implies that the group G is commutative. The commutative group is a group where $a*b = b*a$ for all a, b in G . </think> \boxed{A}

J TRANSFER TO OTHER BENCHMARKS

We ran the Gemma operators, with the directions, probes and thresholds fitted on MMLU, on 300 ARC-Challenge questions (Clark et al., 2018) and 300 GSM8K questions (Cobbe et al., 2021) converted to four-option multiple choice (Appendix D). These runs are exploratory and not part of the protocol. Table 12 shows that the MMLU threshold does not carry to ARC: the post-hoc probe fires on 59 of 300 questions and PTS reaches selectivity 0.100, against 0.370 for the ungated additive dose. On GSM8K PTS keeps 0.944 of confident-right answers at selectivity 0.287, while the additive dose reaches 0.319 at retention 0.634.

method	ARC-Challenge ($n_{OCW} = 66, n_{CR} = 205$)			GSM8K-MCQ ($n_{OCW} = 35, n_{CR} = 161$)		
	Δ_{acc}	retention	Sel	Δ_{acc}	retention	Sel
additive CAA (tuned)	+0.027	0.810	+0.370	+0.023	0.634	+0.319
CAST-style, trace condition	-0.003	0.863	+0.060	-0.060	0.516	+0.173
CAST, prompt condition	-0.017	0.844	+0.223	+0.023	0.634	+0.319
PTS, post-hoc	-0.010	0.888	+0.100	+0.023	0.944	+0.287
PTS, decides at the prompt	-0.013	0.941	-0.013	-0.040	0.814	+0.128

Table 12: Transfer of the Gemma operators fitted on MMLU, M5 readout, 300 questions each, no refitting.

K RESULTS UNDER THE EARLIER PROTOCOL

An earlier version of this work used the verification readout M4 (Appendix L) as its primary instrument, five evaluation subsets of 300 MMLU questions, a detector read off the extraction split, and configurations tuned on a 1,200-question split. We keep its results here as a record, with three qualifications that the confirmatory protocol removes. Under M4 the edited generations are the per-option verification traces, not the reasoning trace, so these results describe the verification task. Four of the five subsets share 7 to 14 questions with the extraction split (38 in total); the pooled numbers below drop them. The subsets also overlap each other in 54 questions, which are counted once.

Pooled comparison. Table 13 pools the five subsets (1,406 unique questions, 245 OCW and 380 CR under M4) and pairs every method with the tuned gated ablation of that protocol. No tuned baseline is above it at the 5% level, and none is below it either except the single-pass gate; the tuned additive dose has the highest mean (+0.271 against +0.203, paired difference +0.067, 95% CI [-0.024, +0.156]). Under M2 on the same traces every selectivity is below 0.11.

method (earlier protocol, tuned)	M4		M2	
	Sel [95% CI]	Δ vs gated abl.	Sel [95% CI]	Δ_{acc}
gated ablation (tuned)	+0.203 [+0.125, +0.282]	—	+0.042 [+0.006, +0.078]	-0.005
additive CAA (tuned)	+0.271 [+0.188, +0.350]	+0.067 [-0.024, +0.156]	+0.082 [+0.039, +0.127]	-0.009
directional ablation	+0.258 [+0.178, +0.338]	+0.054 [-0.017, +0.126]	+0.080 [+0.038, +0.124]	-0.004
CAST-style (tuned)	+0.221 [+0.146, +0.294]	+0.017 [-0.075, +0.108]	+0.101 [+0.060, +0.143]	-0.005
MiMiC (tuned)	+0.126 [+0.050, +0.203]	-0.077 [-0.180, +0.023]	+0.044 [+0.006, +0.082]	-0.006
Linear-AcT (tuned)	+0.146 [+0.082, +0.211]	-0.057 [-0.155, +0.039]	+0.032 [+0.001, +0.064]	-0.004
prompting	+0.196 [+0.118, +0.273]	-0.008 [-0.112, +0.096]	+0.053 [+0.011, +0.096]	-0.011
score-proportional dose (tuned)	+0.241 [+0.163, +0.320]	+0.038 [-0.026, +0.100]	+0.057 [+0.017, +0.099]	-0.008
sequential gate (tuned)	+0.079 [+0.021, +0.138]	-0.124 [-0.215, -0.031]	+0.024 [-0.004, +0.052]	-0.003
gated ablation (published)	+0.278 [+0.201, +0.356]	+0.075 [+0.005, +0.146]	+0.076 [+0.036, +0.117]	+0.004
additive -0.75 (published)	+0.000 [+0.000, +0.000]	-0.203 [-0.282, -0.125]	-0.025 [-0.078, +0.028]	-0.171

Table 13: Earlier protocol, five subsets pooled with overlapping questions removed. Intervals are question-level paired bootstrap intervals (10^4 resamples).

Earlier detector. The two-dimensional LDA gate of the earlier version reached AUROC 0.77–0.80 between OCW and CR on the 400 extraction traces it was fitted to, and 0.59 (layer 14) and 0.59 (layer 16) on the 1,406 pooled evaluation questions; the one-dimensional detection direction reached 0.66. This in-sample optimism is what motivated fitting the detector on a separate pool and reporting only held-out AUROC.

Scale, layers, geometry and cost. Table 14 repeats the published configurations on `gemma-2-9b-it`; Table 15 refits the axes and the gate at nine depths; Table 16 reports displacement, nearest-neighbour distance, the fraction of tokens outside the 99% conditional ellipsoid and next-token KL; Table 17 gives end-to-end wall clock on one H100 for 300 questions.

method (M4, 9B, 3 subsets)	Δacc	ECE	ocw_rm	cr_keep	AURC	Sel
baseline	$\text{acc } 0.633 \pm 0.031$	0.144 ± 0.021	—	—	0.197 ± 0.025	—
additive -0.75	-0.184 ± 0.033	0.167 ± 0.037	1.000 ± 0.000	0.000 ± 0.000	0.450 ± 0.023	$+0.000 \pm 0.000$
CAST-style gate+add	-0.114 ± 0.032	0.101 ± 0.011	0.831 ± 0.058	0.453 ± 0.038	0.285 ± 0.016	$+0.283 \pm 0.093$
clamp q_{50} (ours)	$+0.023 \pm 0.012$	0.143 ± 0.018	0.315 ± 0.012	0.887 ± 0.015	0.184 ± 0.025	$+0.202 \pm 0.020$
gated ablation (ours)	$+0.011 \pm 0.008$	0.127 ± 0.009	0.297 ± 0.026	0.907 ± 0.014	0.186 ± 0.023	$+0.203 \pm 0.034$

Table 14: gemma-2-9b-it, layer 24, published configurations, mean $\pm$ standard deviation over three subsets of 300 questions.

layer	Cohen’s d conf.	Cohen’s d OCW/CR	gate AUROC	2AUC-1
4	+0.525	-0.079	0.783	+0.566
7	+0.535	-0.093	0.756	+0.511
10	+0.512	-0.144	0.741	+0.482
12	+0.516	-0.150	0.742	+0.485
14	+0.589	-0.155	0.769	+0.538
16*	+0.631	-0.141	0.799	+0.599
18	+0.616	-0.116	0.779	+0.558
22	+0.675	-0.110	0.791	+0.582
25	+0.700	-0.111	0.649	+0.298

Table 15: Layer sweep on the 400 extraction traces (in-sample AUROC).

L PROMPTS

All prompts are rendered with the model’s chat template. Text in braces is filled per question; everything else is verbatim.

Reasoning prompt (all readouts).

Given the following multiple choice question, choose the single best answer (A, B, C, or D). Please think step-by-step and put your thinking inside `<think>...</think>` tokens. After `</think>`, give ONLY your final answer in the format: `\boxed{[letter]}`

Question: {question}

Options:

A. {option A}

B. {option B}

C. {option C}

D. {option D}

M5 (primary). After the model’s reply, a second user turn is added to the same conversation and $P(\text{YES})$ is read from the next-token logits, summing the probabilities of the single-token variants of YES and of NO and renormalising over the two:

You answered {letter}. Is that answer correct? Reply with only YES or NO.

The letter is the one with the highest logit after the last `\boxed{}` of the trace, which is appended when the trace has none (6.1% of unsteered confirmatory traces). On the unsteered confirmatory split this letter equals the written letter on 96.9% of questions; the remainder are traces with several boxes or none. A question is confident iff $P(\text{YES}) \geq 0.5$.

M2 (secondary). The confidence of the same generated answer is the softmax over the four letter logits at the position after `\boxed{}`.

M4 (secondary; primary in the earlier version). Each option is verified in a separate generation, with at most 192 new tokens:

Answer whether the given answer is the right answer for the following question. Please think step-by-step and put your thinking inside `<think>...</think>` tokens. After `</think>`, answer ONLY with "YES" or "NO".

operator	MMLU				ARC				GSM8K			
	disp.	NN	out%	KL	disp.	NN	out%	KL	disp.	NN	out%	KL
unsteered	—	1.00	0.9	0.000	—	1.00	1.2	0.000	—	1.00	2.9	0.000
random direction, norm-matched	0.663	1.90	1.6	0.445	0.644	2.27	1.7	0.233	0.749	2.06	0.9	0.149
additive -0.75 Panickssery et al. (2024)	0.663	1.88	100.0	0.032	0.644	2.25	100.0	0.008	0.749	2.04	100.0	0.012
additive, tuned dose Panickssery et al. (2024)	0.221	1.22	69.9	0.005	0.215	1.33	40.3	0.001	—	—	—	—
MiMiC Singh et al. (2024)	0.082	1.06	1.3	0.001	0.074	1.08	1.8	0.000	0.097	1.06	4.5	0.001
Linear-AcT Rodriguez et al. (2025a)	0.012	1.01	1.0	0.000	0.011	1.01	1.5	0.000	0.012	1.01	3.5	0.000
clamp q_{50} (ours)	0.034	1.03	1.3	0.006	0.046	1.06	1.1	0.001	0.003	1.00	2.9	0.001
quantile transport (ours)	0.004	1.00	1.0	0.000	0.003	1.00	1.3	0.000	0.007	1.00	4.0	0.000
ablation Arditì et al. (2024)	0.144	1.15	29.5	0.010	0.178	1.28	28.4	0.001	0.062	1.05	10.1	0.001

Table 16: Activation diagnostics at layer 14 under teacher forcing of unsteered traces. NN is the nearest-neighbour distance to 4,096 natural activations relative to the unsteered value; out% is the share of tokens outside the 99% conditional ellipsoid of the edited coordinate given the leading 64 principal directions of the complement (1% expected); KL is the mean next-token divergence. These are measured distances, not a test of membership in a manifold.

method	passes	seconds	wall clock
unsteered	1	43.5	1.00×
additive -0.75 Panickssery et al. (2024)	1	45.2	1.04×
ablation Arditì et al. (2024)	1	44.6	1.02×
clamp q_{50} (ours)	1	41.4	0.95×
quantile transport (ours)	1	44.8	1.03×
MiMiC Singh et al. (2024)	1	42.9	0.98×
Linear-AcT Rodriguez et al. (2025a)	1	40.9	0.94×
gated ablation, post-hoc gate (ours) (223/300 regenerated)	2	85.8	1.97×
gated ablation, sequential gate (ours)	1	45.5	1.05×

Table 17: End-to-end wall clock for 300 MMLU questions, batch 64, one H100, one run per method.

Question:
{question}
Answer:
{option text}

$P(\text{YES})$ is read at the answer word after `</think>`; the predicted answer is the option with the largest $P(\text{YES})$ and its confidence is that value divided by the sum over options. When an M4 condition is steered, the hook is active in these four verification generations, not in the reasoning trace, so an M4 result describes the verification task.

Prompting baseline. The instruction below is prepended to the user turn (Gemma has no system role) or passed as the system message (Qwen):

Be careful and well-calibrated. Only answer confidently when you are sure; if you are uncertain, explicitly acknowledge the uncertainty in your reasoning and confidence.

Weights. Gemma-2-2B weights were served from the `unsloth/gemma-2-2b-it` mirror, whose 288 tensors are bit-identical to `google/gemma-2-2b-it`; Qwen weights are Qwen/Qwen2.5-7B-Instruct. Decoding is greedy with at most 320 new tokens for the reasoning trace.